

# Book Title

Your Name

2023

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## Preface

In today’s rapidly evolving software landscape, applications rarely exist in isolation. The monolithic architectures of the past are increasingly giving way to dynamic ecosystems of interconnected services. These modern systems often resemble a complex network, pulsing with data, composed of services written in multiple programming languages, deployed and managed by different teams, and crucially, expected to react to events in near real-time. This shift towards distributed, polyglot, and event-driven architectures presents both immense opportunities and significant challenges for development teams.

This book was born from the crucible of such a transformation. It chronicles a practical journey: migrating an established, primarily Go-based system onto the powerful **Apache Kafka** platform. This wasn’t just a simple lift-and-shift operation. It involved strategically introducing **Kotlin** to leverage the strengths of the JVM for specific workloads while ensuring the continued operation and integration of a vital existing **Ruby** service responsible for data science tasks. The goal was not merely to replace one technology with another but to fundamentally reshape the system’s communication patterns, moving away from brittle, tightly-coupled REST calls and overloaded message queues towards a more resilient, scalable, and observable event-driven paradigm.

Along this path, we encountered numerous hurdles and learned valuable lessons. We discovered that simply adopting Kafka wasn’t enough. True success required revisiting fundamental principles of distributed systems and applying them rigorously. We distilled a set of **first principles**—concepts like log-centric thinking, the nuances of loose coupling, the inevitability of eventual consistency, and the practicalities of managing contracts in a polyglot world. Applying these principles allowed us to transform what could have been an overwhelming collection of disparate components into a coherent, understandable, and robust system where events form the central nervous system.

This text is intentionally **opinionated and hands-on**. We believe that practical, working code speaks louder than abstract diagrams or high-level vendor slides. Therefore, you won’t find pseudo-interfaces or theoretical discussions detached from reality. Instead, we focus on showing complete, functioning slices of code, directly extracted from a reference implementation that accompanies this book. We aim to be **light on ceremony**, cutting through the buzzwords to focus on the core concepts and patterns that deliver real-world value. Our goal is to provide a pragmatic guide grounded in production experience.

## Who Is This Book For?

This book is primarily aimed at practitioners who are grappling with the complexities of modern distributed systems, particularly those considering or under-

going a transition to Kafka and event-driven architectures. Specifically, it will be most valuable for:

- **Senior Engineers and Tech Leads:** Individuals responsible for guiding teams through the technical challenges of adopting Kafka, designing event-driven flows, and managing polyglot environments.
- **Developers:** Engineers comfortable working in Go, Ruby, or JVM languages (like Kotlin or Java) who want to understand how different runtimes can effectively integrate and communicate using Kafka as a common backbone. It offers insights into the specific libraries and patterns used in each language.
- **Architects:** Professionals looking for battle-tested, production-oriented patterns and strategies for building scalable, resilient event-driven systems, moving beyond theoretical concepts to see how these ideas are implemented in practice.

While we delve into specific code examples, the underlying principles and patterns discussed are broadly applicable to anyone involved in building or maintaining distributed systems.

## How This Book Is Organized

To guide you through this journey, the book is structured into five main parts, followed by appendices:

- **Part I — Principles:** Lays the conceptual foundation. We explore the transformative power of log-centric thinking, the critical trade-offs involved in achieving loose coupling through eventual consistency, and the reasons why embracing polyglot services is often not just beneficial but necessary in modern systems.
- **Part II — Kafka Essentials:** Dives into the core mechanics of Apache Kafka. We cover topics, partitions, brokers, consumer groups, the practical implications of different delivery semantics, and demystify the often-misunderstood concept of exactly-once processing.
- **Part III — The Polyglot System:** Walks through the practical implementation details of integrating different languages with Kafka. We examine specific libraries and approaches for Go (using Watermill), Kotlin (using Spring Kafka), and Ruby (using `ruby-kafka`), drawing directly from the reference codebase.
- **Part IV — Event-Driven Patterns:** Explores key architectural patterns that are essential for building robust event-driven systems. This includes managing data contracts with schemas (using Protocol Buffers), implementing reliable workflows with Sagas, leveraging Event Sourcing for immutable history, and applying Command Query Responsibility Segregation (CQRS) effectively.
- **Part V — Operations:** Addresses the crucial aspects of running and maintaining an event-driven system in production. We cover observability strategies (logging, tracing, metrics with the ELK stack), comprehensive testing approaches (from contract tests to chaos engineering), deployment considerations (using containers, Docker Compose, and Kubernetes), and strategies for scaling and evolving the system over time.

- **Appendices:** Provide supplementary reference material, including configuration cheat sheets for Kafka clients in different languages, a glossary of key terms, and suggestions for further reading.

## The Reference Implementation

Throughout this book, every code listing, configuration snippet, and command is derived from a fully functional, integration-tested reference implementation. This codebase embodies the principles and patterns discussed, providing a concrete example of the polyglot system built around Kafka. We strongly encourage you to clone the repository, explore the code, and run the system locally using the provided `docker-compose` setup. Following along with the running code is the best way to solidify your understanding and see these concepts in action.

We hope this practical, code-driven exploration of polyglot event-driven systems with Kafka equips you with the knowledge and confidence to build more resilient, scalable, and adaptable software.

## Chapter 1: The Log-centric World-view

In the realm of distributed systems, few abstractions have proven as powerful and transformative as the humble log. Despite its simplicity—or perhaps because of it—the log has emerged as a foundational concept that underpins many of today’s most scalable and resilient systems. This chapter explores how adopting a log-centric perspective fundamentally changes how we design, build, and reason about distributed applications, particularly when using Apache Kafka.

### Beyond Simple Logging: The Distributed Log

When most developers hear the term “log,” they immediately think of application logs—those streams of text messages that help debug issues or trace program execution. While useful, these text-based logs represent only a narrow application of a much more powerful concept.

A **distributed log** is not merely a queue with the ability to seek backward. It represents something far more profound: it is the **memory** of your system. In a log-centric architecture, the log becomes the authoritative record of everything that has happened within your system—the single source of truth from which all other state can be derived.

At its core, a log is an append-only, totally-ordered sequence of records ordered by time. Each entry is assigned a unique, sequential identifier (often called an offset or sequence number) that establishes its position in the log. This structure is deceptively simple yet incredibly powerful:

```
[0] Order #1234 created for customer C001
[1] Payment of $59.99 received for order #1234
[2] Inventory item SKU-789 reserved for order #1234
[3] Shipping label created for order #1234
[4] Order #1234 marked as shipped
```

In this example, each log entry represents an event that occurred in the system. The numbers in brackets represent the log offsets—immutable positions that define the strict ordering of events. This ordering creates a notion of “time” that is decoupled from physical clocks, providing a consistent view of causality across distributed components.

## The Inversion of Data Flow

Traditional systems often rely on mutable databases as their source of truth. Applications query these databases to retrieve the current state, modify that state, and write it back. This approach, while familiar, creates numerous challenges in distributed environments: contention, locking, consistency issues across services, and difficulties in tracking how state evolved over time.

The log-centric approach inverts this model. Instead of treating the current state as primary and changes as transient operations, it treats the **changes themselves** as the primary record. Everything that happens—an order submission, an inventory check, a payment rejection—is persisted as an **immutable, ordered event** in the log. The current state becomes a derived artifact, a point-in-time projection created by processing the log from the beginning (or from a known checkpoint) up to the present.

This inversion has profound implications:

### 1. Rebuilding State is Cheap; Snapshots are an Optimization

Since the log contains the complete history of changes, any system can reconstruct the full state by replaying events from the beginning. This property makes recovery from failures straightforward—a crashed service simply reprocesses the log to rebuild its state.

Of course, for efficiency, systems often maintain snapshots or checkpoints to avoid replaying the entire log. But importantly, these snapshots are merely optimizations, not the source of truth. The log remains the authoritative record.

```
// Simplified example of state reconstruction from events
func rebuildOrderState(events []Event) Order {
    var order Order

    for _, event := range events {
        switch event.Type {
        case "OrderCreated":
            order = Order{ID: event.OrderID, Status: "Created"}
        case "PaymentReceived":
            order.Status = "Paid"
        case "InventoryReserved":
            order.Status = "Preparing"
        case "Shipped":
```

# Chapter 2: Loose Coupling & Eventual Consistency

In distributed systems, particularly those built around microservices, two interrelated co

## ## The Problem with Tight Coupling

Traditional service-to-service communication often relies on direct, synchronous calls. In  
Service A → HTTP Request → Service B ← HTTP Response ←

This pattern is intuitive and familiar, mirroring how we naturally think about interaction

### ### Availability Coupling

When Service A depends directly on Service B, it inherits Service B's availability character

Consider an e-commerce platform where an order service directly calls a payment service to

### ### Temporal Coupling

Synchronous communication patterns create temporal coupling-  
a dependency on both services being available and responsive at the same moment in time. S

### ### Knowledge Coupling

Direct calls often require the caller to have detailed knowledge about the callee's API, d

### ### Scale Coupling

When services communicate synchronously, they must scale together to handle peak loads. If  
even if the increased load is temporary or specific to Service A's domain.

## ## Loose Coupling Through Event-Driven Architecture

Event-driven architecture with Apache Kafka offers an alternative approach that addresses

Service A → Publish Event → Kafka Topic → Consume Event → Service B

This fundamental shift in communication pattern enables loose coupling in several dimension

### ### Decoupling Availability

In an event-driven topology, Service A only needs Kafka to be available when publishing ev  
it doesn't depend directly on Service B's availability. If Service B is temporarily unavail

Returning to our e-commerce example, the order service would publish an `OrderCreated` eve

### ### Decoupling Time

Asynchronous communication removes temporal coupling. Service A can publish an event and i

### ### Decoupling Knowledge

With event-driven communication, services need only understand the events they produce and

### ### Decoupling Scale

Services can scale independently based on their specific workloads. If Service A experienc

### ## The Trade-off: Eventual Consistency

The benefits of loose coupling through event-driven architecture come with a significant t

### ### From ACID to BASE

Traditional monolithic applications often rely on ACID transactions (Atomicity, Consistenc

In distributed, event-driven systems, we typically embrace BASE properties instead (Basica

- **Basically Available**: The system remains operational even when parts of it are unavail
- **Soft state**: The system's state may change over time, even without input, due to even
- **Eventually consistent**: The system will become consistent over time, given that the s

### ### Understanding Eventual Consistency

Eventual consistency means that if no new updates are made to a given data item, eventuall

In our e-commerce example, after a customer places an order, they might see it in a "Proce  
a window of eventual consistency.

Time 0: Customer places order → OrderCreated event published  
Time 1: Order service shows status "Processing"  
Time 2: Payment service processes payment → PaymentCompleted event published  
Time 3: Order service consumes PaymentCompleted event, updates status to "Paid"  
Time 4: Customer sees updated "Paid" status

During the window between Time 1 and Time 4, the system is in an inconsistent state-  
different services have different views of the order's status. This inconsistency is tempo

### ### Designing for Eventual Consistency

Embracing eventual consistency requires explicit design considerations:

#### #### 1. Idempotent Handlers

Since Kafka guarantees "at least once" delivery by default, services must be prepared to r

```
```go
// Example of an idempotent handler in Go
func handlePaymentCompleted(event PaymentCompletedEvent) error {
    // Use a unique key from the event to detect duplicates
    key := fmt.Sprintf("payment:%s", event.PaymentID)
```



```

    // Check if we've already processed this payment
    processed, err := redisClient.SetNX(key, "processed", 24*time.Hour).Result()
    if err != nil {
        return err
    }

    // If already processed, acknowledge but do nothing
    if !processed {
        log.Printf("Payment %s already processed, skipping", event.PaymentID)
        return nil
    }

    // Process the payment completion
    return updateOrderStatus(event.OrderID, "Paid")
}

```

**2. Compensation Logic** Since we can't rely on atomic transactions across services, we need explicit compensation logic to handle failures. If a step in a multi-service process fails, we need to publish events that trigger compensating actions in services that have already processed earlier steps.

```

// Example of compensation logic in Kotlin
@KafkaListener(topics = ["inventory.reservation.failed"])
fun handleInventoryReservationFailed(event: InventoryReservationFailedEvent) {
    // Find the payment we previously processed
    val payment = paymentRepository.findByOrderId(event.orderId)

    // Initiate a refund as compensation
    if (payment != null) {
        paymentService.refundPayment(payment.paymentId)

        // Publish an event about the refund
        val refundEvent = PaymentRefundedEvent(
            orderId = event.orderId,
            paymentId = payment.paymentId,
            amount = payment.amount,
            reason = "Inventory reservation failed"
        )
        kafkaTemplate.send("payment.refunded", refundEvent)
    }
}

```

**3. Explicit State Transitions** Services should maintain explicit state machines that track the progress of business processes. These state machines define valid state transitions and ensure that events are processed in a way that maintains business invariants despite eventual consistency.

```

# Example of explicit state transitions in Ruby
class Order
    STATES = %w[created payment_pending paid preparing shipped delivered canceled]

```

```

def transition_to(new_state, metadata = {})
  return false unless STATES.include?(new_state)

  case state
  when "created"
    return false unless ["payment_pending", "canceled"].include?(new_state)
  when "payment_pending"
    return false unless ["paid", "canceled"].include?(new_state)
  when "paid"
    return false unless ["preparing", "canceled"].include?(new_state)
  when "preparing"
    return false unless ["shipped", "canceled"].include?(new_state)
  when "shipped"
    return false unless ["delivered"].include?(new_state)
  when "delivered", "canceled"
    return false # Terminal states
  end

  # If we get here, the transition is valid
  self.state = new_state
  self.state_changed_at = Time.now
  self.state_metadata = metadata
  save
end
end

```

**4. User Experience Considerations** Eventual consistency has implications for user experience. Applications need to set appropriate expectations and provide feedback that acknowledges the asynchronous nature of operations.

For example, after a user places an order, instead of showing “Order Confirmed” immediately, the UI might show “Order Received - Processing Payment” until the payment confirmation event is processed. Progress indicators, status updates, and clear messaging help users understand the system’s behavior.

## Formalizing the Trade-offs: Patterns for Eventual Consistency

Several established patterns help formalize and manage the trade-offs inherent in eventually consistent systems:

### The Saga Pattern

A saga is a sequence of local transactions where each transaction updates data within a single service. The saga ensures consistency across services by using compensating transactions to undo changes when something fails.

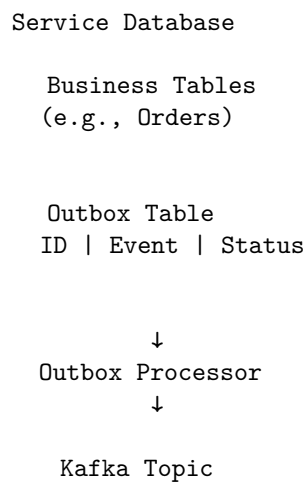
Order Service: Create Order → Payment Service: Process Payment → Inventory Service: Reserve

Order Service: Cancel Order ← Payment Service: Refund Payment ← Inventory Service: Release

We'll explore sagas in depth in Chapter 11, but they represent one of the most important patterns for managing complex workflows in eventually consistent systems.

### The Outbox Pattern

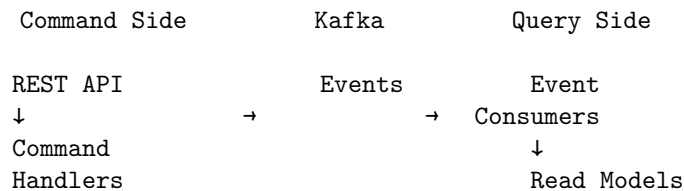
The outbox pattern ensures reliable event publishing by storing events in a local “outbox” table within the same transaction that updates the service’s main database. A separate process then reads from this outbox and publishes the events to Kafka.



This pattern bridges the gap between local ACID transactions and distributed eventual consistency, ensuring that database updates and event publishing are effectively atomic.

### CQRS (Command Query Responsibility Segregation)

CQRS separates the write model (commands) from the read model (queries), allowing each to be optimized independently. In an eventually consistent system, commands produce events that are used to update read models asynchronously.



We'll explore CQRS in detail in Chapter 13, but it's worth noting here as a pattern that embraces and leverages eventual consistency rather than fighting against it.

## Practical Considerations for Loose Coupling

While the theoretical benefits of loose coupling are compelling, implementing it effectively requires attention to several practical considerations:

### Topic Design

The design of Kafka topics significantly impacts the degree of coupling between services. Topics should be organized around business domains and events rather than services, promoting a domain-driven design approach.

For example, instead of a topic named `payment-service-output`, prefer `payments.processed` or `orders.payment-status-changed`, focusing on the business event rather than the service that produced it.

### Schema Evolution

As services evolve, so do the events they produce and consume. A well-designed schema evolution strategy is essential for maintaining loose coupling over time. This typically involves using schema registries, versioning, and backward/forward compatibility rules.

We'll explore this topic in depth in Chapter 10 when discussing contracts and schemas with Protocol Buffers.

### Monitoring and Observability

Eventual consistency introduces challenges for monitoring and debugging. Traditional request-response patterns provide immediate feedback on success or failure, while event-driven systems require more sophisticated observability solutions.

Correlation IDs, distributed tracing, and event logging are essential tools for understanding the flow of events and diagnosing issues in loosely coupled systems. We'll cover these topics in Chapter 14 on observability.

## Conclusion

Loose coupling and eventual consistency represent fundamental trade-offs in distributed system design. By embracing event-driven architecture with Apache Kafka, we gain significant benefits in terms of resilience, scalability, and service independence, at the cost of immediate consistency.

This trade-off isn't suitable for every use case—some scenarios genuinely require immediate consistency and tight coupling. However, for many business domains, particularly those with natural asynchronous processes (like order fulfillment, content publishing, or data processing), the benefits of loose coupling outweigh the costs.

As we'll see in subsequent chapters, patterns like sagas, event sourcing, and CQRS provide structured approaches to managing the challenges of eventual consistency while leveraging the benefits of loose coupling. By understanding these trade-offs and applying appropriate patterns, we can build systems that

are both resilient and maintainable, capable of evolving with changing business needs.

## Chapter 3: Why Polyglot Services are Inevitable

In the evolution of software architecture, particularly in the realm of microservices, a significant trend has emerged: the adoption of polyglot approaches. Rather than standardizing on a single technology stack across an entire system, organizations increasingly embrace multiple programming languages, frameworks, and data stores. This chapter explores why this diversity isn't merely a matter of preference or coincidence, but rather an inevitable and often beneficial outcome of modern software development practices—especially when facilitated by an event-driven backbone like Apache Kafka.

### The Monolith Mindset: One Stack to Rule Them All

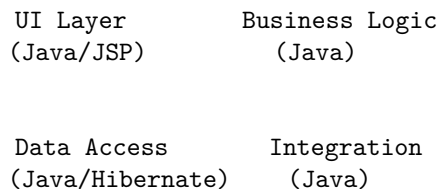
Traditionally, enterprise systems were built as monoliths using a standardized technology stack. This approach offered several advantages:

- **Consistency:** Developers could apply the same patterns and practices throughout the codebase
- **Knowledge sharing:** Team members could easily understand and contribute to any part of the system
- **Simplified operations:** A single deployment unit with uniform monitoring, logging, and scaling approaches
- **Streamlined hiring:** Recruiting focused on expertise in a specific stack (e.g., Java/Spring/Oracle)

This uniformity was often mandated through architectural governance and technology standardization committees. The underlying assumption was that consistency would lead to efficiency and maintainability.

In a monolithic architecture, this assumption largely held true. When all components run within the same process and deployment unit, they naturally share the same runtime environment. Using multiple languages within a monolith would introduce unnecessary complexity with few benefits.

#### Monolithic Application



↓

Database  
(Oracle/PostgreSQL)

However, as systems grow in complexity and organizations adopt microservices architectures, the calculus changes dramatically.

## The Microservices Catalyst: Bounded Contexts and Team Autonomy

The transition to microservices introduces two fundamental shifts that make polyglot approaches not just possible, but advantageous:

### 1. Bounded Contexts and Domain-Driven Design

Microservices architecture often aligns with Domain-Driven Design (DDD) principles, particularly the concept of bounded contexts. Each microservice encapsulates a specific business domain or capability, with its own data model, business rules, and interfaces.

These bounded contexts naturally have different characteristics and requirements:

- **Order processing** might prioritize transactional integrity and consistency
- **Search functionality** might emphasize query flexibility and performance
- **Analytics** might focus on data processing capabilities and mathematical libraries
- **User interfaces** might value rendering performance and frontend frameworks

When services are separated by clear boundaries, they can be optimized independently for their specific requirements, including the choice of programming language and data storage.

### 2. Team Autonomy and Conway’s Law

Conway’s Law observes that “organizations design systems that mirror their communication structure.” In microservices environments, this often manifests as autonomous teams responsible for specific services or domains.

These teams typically have:

- **End-to-end ownership** of their services, from design to deployment and operations
- **Decision-making authority** over implementation details, including technology choices
- **Different backgrounds and expertise** that influence their technology preferences

When teams have autonomy, they naturally gravitate toward technologies that:

- Match their expertise and experience
- Solve their specific domain problems effectively
- Enable them to deliver value quickly and reliably

This team-based decision-making inevitably leads to technology diversity across the organization.

## The Pragmatic Reality: Different Tools for Different Jobs

Beyond organizational factors, there's a simple pragmatic reality: different programming languages and frameworks excel at different tasks. In our reference implementation, we've deliberately chosen three languages—Go, Kotlin, and Ruby—each for specific strengths that align with particular service requirements.

### Go: Efficiency and Simplicity

Go (Golang) has emerged as a popular language for microservices, particularly those with high performance requirements or system-level interactions. Its strengths include:

- **Efficient resource utilization:** Go's lightweight goroutines and small memory footprint make it ideal for services that need to handle many concurrent connections with minimal resources
- **Fast startup time:** Go compiles to static binaries that start almost instantly, beneficial for containerized environments and auto-scaling scenarios
- **Simplicity and readability:** Go's minimalist design philosophy reduces cognitive load and makes services easier to maintain
- **Strong standard library:** Particularly for networking, HTTP handling, and concurrency

In our reference implementation, we use Go for core services that handle high-throughput event processing and require efficient resource utilization:

```
// Example from our Order Service in Go
func main() {
    router, err := watermill.NewRouter(watermill.RouterConfig{})
    if err != nil {
        log.Fatal(err)
    }

    kafkaSubscriber, err := kafka.NewSubscriber(
        kafka.SubscriberConfig{
            Brokers: []string{"kafka:9092"},
            ConsumerGroup: "order-service",
        },
        watermill.NewStdLogger(false, false),
    )
    if err != nil {
        log.Fatal(err)
    }

    // Handler for processing order events
    router.AddHandler(
        "order.created.handler",
```

```

        "order.created",
        kafkaSubscriber,
        "payment.requested",
        kafkaPublisher,
        handlers.HandleOrderCreated,
    )

    // Start the router
    ctx := context.Background()
    go func() {
        err := router.Run(ctx)
        if err != nil {
            log.Fatal(err)
        }
    }()

    // HTTP API for receiving orders
    http.HandleFunc("/orders", handleOrderSubmission)
    log.Fatal(http.ListenAndServe(":8080", nil))
}

```

This Go service efficiently handles both HTTP requests and Kafka event processing with minimal resource overhead, making it ideal for high-throughput scenarios.

## Kotlin: JVM Ecosystem with Modern Features

Kotlin represents the modern face of JVM languages, offering significant advantages over traditional Java while maintaining full interoperability with Java libraries and frameworks:

- **Concise syntax:** Kotlin reduces boilerplate compared to Java, increasing developer productivity
- **Null safety:** Built-in null safety features help prevent the infamous `NullPointerException`
- **Coroutines:** First-class support for asynchronous programming without the complexity of callbacks or reactive streams
- **JVM ecosystem access:** Leverages the vast Java ecosystem of libraries, frameworks, and tools
- **Interoperability:** Seamless integration with existing Java code and libraries

In our reference implementation, we use Kotlin with Spring Kafka for services that benefit from the rich JVM ecosystem, particularly for complex business logic and integration with enterprise systems:

```

// Example from our Shipping Service in Kotlin
@Service
class ShippingService(
    private val kafkaTemplate: KafkaTemplate<String, ByteArray>,
    private val shipmentRepository: ShipmentRepository
) {

```



```

@KafkaListener(topics = ["order.confirmed"])
fun processConfirmedOrder(eventBytes: ByteArray) {
    val event = OrderConfirmedEvent.parseFrom(eventBytes)

    // Create shipment record
    val shipment = Shipment(
        orderId = event.orderId,
        customerId = event.customerId,
        items = event.itemsList.map { ShipmentItem(it.sku, it.quantity) },
        status = ShipmentStatus.PENDING
    )

    // Save to database
    shipmentRepository.save(shipment)

    // Initiate shipping process asynchronously
    initiateShipping(shipment)

    // Publish event about shipment creation
    val shipmentCreatedEvent = ShipmentCreatedEvent.newBuilder()
        .setShipmentId(shipment.id)
        .setOrderId(shipment.orderId)
        .setStatus(ShipmentStatus.PENDING.name)
        .build()

    kafkaTemplate.send("shipment.created", shipmentCreatedEvent.toByteArray())
}

@Async
fun initiateShipping(shipment: Shipment) = runBlocking {
    // Complex shipping logic with external integrations
    // Leveraging Kotlin coroutines for asynchronous operations
    val labelJob = async { generateShippingLabel(shipment) }
    val carrierJob = async { selectOptimalCarrier(shipment) }

    val label = labelJob.await()
    val carrier = carrierJob.await()

    // Update shipment with details
    shipment.trackingNumber = label.trackingNumber
    shipment.carrier = carrier.name
    shipment.status = ShipmentStatus.PROCESSING
    shipmentRepository.save(shipment)

    // Publish updated status
    publishShipmentUpdate(shipment)
}
}

```

This Kotlin service leverages Spring's dependency injection, JPA for database

access, and Kotlin’s coroutines for asynchronous operations—all while maintaining clean, concise code.

## Ruby: Rapid Development and Data Processing

Ruby continues to excel in scenarios where development speed, readability, and data manipulation capabilities are more important than raw performance:

- **Expressive syntax:** Ruby’s elegant, readable syntax optimizes for developer productivity
- **Rich ecosystem:** Gems like ActiveRecord, Sinatra, and ruby-kafka provide powerful abstractions
- **Data processing:** Excellent for data transformation, analysis, and reporting
- **Rapid prototyping:** Enables quick iteration and experimentation
- **Domain-specific languages:** Makes it easy to create expressive APIs for specific domains

In our reference implementation, we use Ruby for analytics and reporting services where its data processing capabilities shine:

```
# Example from our Analytics Service in Ruby
require 'kafka'
require 'json'
require 'active_record'

# Configure Kafka client
kafka = Kafka.new(
  seed_brokers: ["kafka:9092"],
  client_id: "analytics-service"
)

# Set up consumer
consumer = kafka.consumer(group_id: "analytics-group")
consumer.subscribe("order.created")
consumer.subscribe("payment.processed")
consumer.subscribe("shipment.delivered")

# Process messages
consumer.each_message do |message|
  case message.topic
  when "order.created"
    event = Ruby::IDL::OrderCreatedEvent.decode(message.value)
    OrderAnalytics.record_new_order(
      order_id: event.order_id,
      customer_id: event.customer_id,
      total_amount: event.total_amount,
      item_count: event.items.size
    )
  when "payment.processed"
```

```

    event = Ruby::IDL::PaymentProcessedEvent.decode(message.value)
    OrderAnalytics.record_payment(
      order_id: event.order_id,
      payment_method: event.payment_method,
      amount: event.amount,
      success: event.success
    )

  when "shipment.delivered"
    event = Ruby::IDL::ShipmentDeliveredEvent.decode(message.value)
    OrderAnalytics.record_delivery(
      order_id: event.order_id,
      delivery_time: Time.at(event.delivery_timestamp),
      shipping_duration: calculate_shipping_duration(event)
    )
  end
end

```

This Ruby service efficiently processes events from multiple topics, transforming them into analytics records with minimal code. The expressive syntax makes the business logic clear and maintainable.

## Kafka: The Polyglot Enabler

While polyglot architectures offer significant benefits, they also introduce integration challenges. Different languages have different idioms, libraries, and approaches to common problems. Without a unifying backbone, these differences can lead to brittle, complex integration points.

Apache Kafka serves as the ideal backbone for polyglot architectures for several reasons:

### 1. Language-Agnostic Protocol

Kafka's protocol is language-agnostic, with client libraries available for virtually every programming language. This means services can interact with Kafka using native libraries that follow the idioms and patterns of their respective languages.

### 2. Decoupled Communication

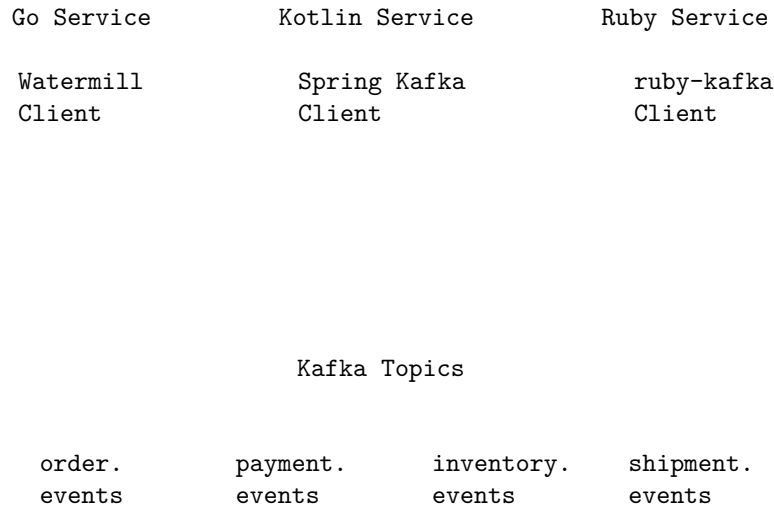
By using Kafka as an intermediary, services don't need to know about each other's implementation details. They communicate through well-defined events rather than direct API calls, reducing the coupling between different technology stacks.

### 3. Schema Management

With tools like Protocol Buffers (which we'll explore in Chapter 10), Kafka enables strict schema definitions that can be used to generate code in multiple languages. This ensures that events are consistently structured and interpreted across language boundaries.

#### 4. Unified Truth Stream

Kafka provides a single, unified stream of truth that all services can access, regardless of their implementation language. This shared event log becomes the common language that bridges the gaps between different technology stacks.



In this architecture, each service interacts with Kafka using client libraries that feel natural in their respective languages, while the events themselves provide a common language for cross-service communication.

### Managing Polyglot Complexity

While polyglot architectures offer significant benefits, they also introduce challenges that must be managed:

#### 1. Operational Complexity

Each technology stack brings its own operational considerations: deployment patterns, monitoring approaches, debugging tools, and performance characteristics. This diversity can increase the operational burden on teams.

To mitigate this challenge:

- **Containerization:** Use Docker to standardize deployment regardless of language
- **Unified observability:** Implement consistent logging, metrics, and tracing across all services
- **Infrastructure as code:** Automate environment setup to ensure consistency

## 2. Knowledge Silos

When different services use different technologies, there's a risk of knowledge silos forming, where only certain team members can work on specific services.

To address this:

- **Cross-training:** Encourage developers to learn multiple languages and technologies
- **Pair programming:** Pair experts in different technologies to share knowledge
- **Internal documentation:** Maintain clear documentation on each service's implementation

## 3. Proliferation Control

Without governance, technology choices can proliferate uncontrollably, leading to a fragmented landscape that's difficult to maintain.

Effective strategies include:

- **Curated diversity:** Limit the organization to a manageable set of approved technologies
- **Justification process:** Require clear business justification for introducing new technologies
- **Technology radar:** Regularly evaluate and categorize technologies (adopt, trial, assess, hold)

## The Balanced Approach: Strategic Polyglotism

Rather than embracing polyglot architectures indiscriminately or rejecting them entirely, organizations should adopt a strategic approach:

1. **Identify domain-specific requirements:** Understand the unique characteristics and requirements of each bounded context
2. **Evaluate technology fit:** Select technologies based on their alignment with these requirements
3. **Consider team capabilities:** Factor in the expertise and preferences of the teams responsible for each service
4. **Establish boundaries:** Define clear contracts and communication patterns between services using different technologies
5. **Implement shared infrastructure:** Provide common platforms for deployment, monitoring, and integration

This balanced approach allows organizations to leverage the benefits of polyglot architectures while managing their complexity.

## Conclusion

The polyglot nature of modern microservices architectures isn't a trend or a coincidence—it's an inevitable outcome of the forces at play in contemporary software development. When services are separated by clear boundaries, owned by autonomous teams, and optimized for specific domains, technology diversity naturally emerges.

Apache Kafka serves as the ideal backbone for these polyglot systems, providing a language-agnostic communication layer that allows services to interact through well-defined events rather than brittle API calls. This event-driven approach enables each service to use the technology stack best suited to its specific requirements while maintaining a cohesive overall system.

As we'll explore in subsequent chapters, particularly in Part III, each language brings its own strengths and idioms to Kafka integration. Go offers efficiency and simplicity with libraries like Watermill, Kotlin provides rich ecosystem access with Spring Kafka, and Ruby enables rapid development with `ruby-kafka`. By understanding these strengths and leveraging Kafka as a unifying backbone, we can build systems that are both technologically diverse and cohesively integrated—systems that truly represent the best of polyglot architecture.

## Chapter 4: Kafka From 10,000 ft (Enhanced)

Apache Kafka has emerged as one of the most powerful and versatile distributed streaming platforms, serving as the backbone for event-driven architectures across industries. Before diving into the implementation details of our polyglot system, it's essential to understand Kafka's core concepts and architecture. This chapter provides a high-level overview of Kafka, explaining its key components and how they work together to enable scalable, resilient event streaming, drawing insights from foundational texts like “Kafka in Action” by Scott, Gamov, and Klein.

### The Evolution of Kafka

Apache Kafka was originally developed at LinkedIn around 2010 to address the growing need for real-time data processing at scale. Traditional message queues and enterprise service buses couldn't handle LinkedIn's requirements for high throughput, fault tolerance, and horizontal scalability. The team, led by Jay Kreps (who later founded Confluent), designed Kafka as a distributed commit log that could serve as a central data pipeline for the entire organization.

Since its inception, Kafka has evolved from a messaging system to a comprehensive event streaming platform. It now encompasses not only the core broker functionality but also Kafka Connect (for integrating with external systems), Kafka Streams (for stream processing), and `ksqlDB` (for SQL-based stream processing). It's used by thousands of companies worldwide, from startups to Fortune 500 enterprises, for use cases ranging from real-time analytics and data integration to microservices communication and stream processing.

### Core Components of Kafka

At its heart, Kafka is a distributed, partitioned, replicated commit log service. Let's break down the key components that make up a Kafka deployment:

#### Topics: The Fundamental Organizing Unit

A **topic** is a named, append-only log to which data can be published. Topics serve as the primary categorization mechanism in Kafka, typically repre-

senting a particular stream of data or class of events (e.g., `order.events`, `user.activity`).

Topics are logical entities that are physically implemented as partitions distributed across the Kafka cluster. They provide the abstraction that allows producers and consumers to work with streams of related events without worrying about the underlying distribution.

### Partitions: Enabling Parallelism and Scalability

Each topic is divided into one or more **partitions**, which are the actual append-only logs that store the data. Partitions serve several critical purposes:

1. **Horizontal Scaling:** By splitting a topic across multiple partitions, Kafka can distribute the load across multiple brokers, enabling horizontal scalability.
2. **Parallelism:** Multiple consumers can process different partitions of the same topic simultaneously, increasing throughput.
3. **Ordering Guarantees:** Within a single partition, Kafka guarantees that messages are stored and delivered in the exact order they were received. Ordering is *not* guaranteed across partitions of the same topic.

Each message within a partition is assigned a sequential ID called an **offset**, which uniquely identifies its position in the partition. This offset serves as a “cursor” that consumers use to track their progress through the partition. Offsets are immutable and monotonically increasing within a partition.

Topic: `order.created`

| Partition 0 | Partition 1 | Partition 2 |
|-------------|-------------|-------------|
| Offset: 0   | Offset: 0   | Offset: 0   |
| Offset: 1   | Offset: 1   | Offset: 1   |
| Offset: 2   | Offset: 2   | Offset: 2   |
| Offset: 3   | Offset: 3   | Offset: 3   |
| ...         | ...         | ...         |

When publishing messages to a topic, producers can specify a partition key. Messages with the same key will always go to the same partition, ensuring that related events are processed in order. If no key is specified, the producer typically uses a round-robin strategy to distribute messages across partitions. For example, in our order processing system, we use the order ID as the partition key to ensure that all events related to a specific order are processed sequentially.

### Brokers: The Servers That Power Kafka

A Kafka cluster consists of one or more **brokers**—servers that store the partitions and serve client requests. Each broker hosts a subset of the partitions across all topics, distributing the storage and processing load. Brokers are identified by an integer ID and can be added to or removed from a cluster dynamically. When a new broker joins the cluster, partitions are automatically redistributed

to balance the load (though this might require manual intervention or specific tooling depending on the Kafka version and configuration).

## Chapter 5: Breaking Down Topics & Partitions (Enhanced)

In the previous chapter, we introduced Apache Kafka from a high-level perspective, covering its core components and architecture. Now, we delve deeper into two of the most fundamental concepts: topics and partitions. Understanding how to design, configure, and manage topics and partitions effectively is crucial for building scalable, resilient, and maintainable Kafka-based systems. This chapter explores the nuances of topic design, the intricacies of partitioning, and the dynamics of consumer groups and rebalancing, drawing on insights from “Kafka in Action” and “Kafka Streams in Action.”

### Designing Effective Topics

Topics are the primary organizational unit in Kafka, representing streams of related events. While seemingly simple, the design of your topics significantly impacts system architecture, performance, and maintainability.

#### Naming Conventions

Establishing clear and consistent naming conventions for topics is essential for managing a growing Kafka deployment. Good naming conventions make it easier to understand the purpose of a topic, manage access control, and configure monitoring and alerting.

Common patterns include:

- **Domain-driven naming:** `domain.event_type` (e.g., `orders.created`, `payments.processed`, `inventory.updated`). This aligns topics with business domains and events, promoting loose coupling.
- **Hierarchical naming:** Using dots (.) or underscores (\_) to create namespaces (e.g., `production.ecommerce.orders.v1`).
- **Including versioning:** Explicitly versioning topics (e.g., `orders.created.v2`) can help manage schema evolution, although schema registries often provide more robust solutions.
- **Environment prefixes:** Including environment information (e.g., `dev.orders.created`, `prod.orders.created`) to distinguish between development, testing, and production environments.

In our reference implementation, we primarily use domain-driven naming, focusing on the business event being represented (e.g., `order.created`, `payment.processed`).

As Scott, Gamov, and Klein note in “Kafka in Action,” consistent naming conventions become increasingly important as your Kafka ecosystem grows: “A well-thought-out naming convention will save you countless hours of confusion and debugging as your Kafka deployment scales.”



## Topic Granularity: Single vs. Multiple Topics

A key design decision is the granularity of your topics. Should you use a single topic for all events related to a domain, or multiple topics for different event types?

**Single Topic per Domain** Using a single topic (e.g., `orders`) for all order-related events (`OrderCreated`, `OrderUpdated`, `OrderShipped`):

**Advantages:** - Simplifies producer configuration - Maintains ordering across all event types for the same entity (e.g., all events for Order #1234) - Reduces the number of topics to manage - Simplifies consumer code when processing multiple event types together

**Disadvantages:** - Requires consumers to filter events they aren't interested in - May lead to unnecessary data transfer if consumers only need specific event types - Can complicate schema evolution if different event types evolve at different rates

**Multiple Topics per Event Type** Using separate topics for each event type (e.g., `orders.created`, `orders.updated`, `orders.shipped`):

**Advantages:** - Provides clearer separation of concerns - Allows consumers to subscribe only to the specific events they need - Simplifies schema evolution for individual event types - Enables more granular access control and monitoring

**Disadvantages:** - Increases the number of topics to manage - Complicates maintaining order across different event types for the same entity - May require consumers to join data from multiple topics

```
// Single Topic Approach
Topic: orders
Events: OrderCreated, OrderUpdated, OrderShipped
```

```
// Multiple Topic Approach
Topic: orders.created
Events: OrderCreated
```

```
Topic: orders.updated
Events: OrderUpdated
```

```
Topic: orders.shipped
Events: OrderShipped
```

The best approach depends on the specific use case, the relationships between events, and the access patterns of consumers. Often, a hybrid approach works well, grouping closely related events into a single topic while separating distinct event streams.

Bejeck emphasizes in “Kafka Streams in Action” that “topic design should be driven by how the data will be consumed, not just how it’s produced.” Consider the downstream consumers and their needs when designing your topic structure.

## Topic Configuration Parameters

Beyond naming and granularity, several key configuration parameters affect topic behavior:

**Number of Partitions (`num.partitions`)** The number of partitions for a topic is a critical configuration parameter that impacts scalability, parallelism, and ordering guarantees. Choosing the right number of partitions requires balancing several factors:

- **Throughput:** More partitions generally allow for higher producer and consumer throughput, as the load can be distributed across more brokers and consumer instances.
- **Parallelism:** The maximum parallelism for a consumer group is limited by the number of partitions. If you have 10 partitions, you can run at most 10 consumer instances in parallel within a single group.
- **Ordering:** Kafka only guarantees message order within a single partition. If strict ordering is required for related events, they must be sent to the same partition.
- **Resource Utilization:** Each partition consumes resources on the brokers (file handles, memory, CPU). Too many partitions can increase broker load and potentially impact latency.
- **Rebalancing Time:** More partitions can lead to longer consumer group rebalance times.

From “Kafka in Action,” a general formula for estimating the minimum number of partitions needed for a topic:

$$\text{min\_partitions} = \max(\text{throughput\_in} / \text{producer\_throughput}, \text{throughput\_out} / \text{consumer\_throughput})$$

Where: - `throughput_in` is the expected peak write throughput for the topic - `producer_throughput` is the throughput a single producer can achieve - `throughput_out` is the expected peak read throughput - `consumer_throughput` is the throughput a single consumer can achieve

For example, if you expect a peak write throughput of 100MB/s, each producer can handle 10MB/s, and each consumer can process 5MB/s, you would need at least  $\max(100/10, 100/5) = \max(10, 20) = 20$  partitions.

In practice, it's common to add a buffer (e.g., 30-50%) to account for future growth and throughput variations.

**Replication Factor (`replication.factor`)** The replication factor determines how many copies of each partition are maintained across the cluster:

- **Durability:** Higher replication factors provide better durability in case of broker failures.
- **Availability:** With more replicas, the system can tolerate more broker failures while remaining available.
- **Resource Usage:** Each replica consumes disk space and network bandwidth.

Common settings: - **Development/Testing:** 1 (no replication) - **Production (minimum):** 3 (can tolerate 2 broker failures) - **Critical Production:** 5 (can

tolerate 4 broker failures)

The formula for the number of broker failures a system can tolerate is:

```
max_failures = replication_factor - 1
```

### Retention Settings

- **Time-based retention** (`retention.ms`): How long messages are kept before deletion.
- **Size-based retention** (`retention.bytes`): Maximum size a partition can grow to before old segments are deleted.
- **Cleanup policy** (`cleanup.policy`):
  - `delete`: Standard deletion based on retention settings
  - `compact`: Log compaction (keep only the latest value for each key)
  - `compact,delete`: Combination of both (compact but also enforce time/size limits)

### Segment Settings

- **Segment size** (`segment.bytes`): Maximum size of a segment file before a new one is created.
- **Segment time** (`segment.ms`): Maximum time before a new segment is created.

Smaller segments allow for more granular retention but create more files to manage.

## The Power of Partitioning

Partitioning is Kafka's mechanism for achieving horizontal scalability and parallelism. Understanding how partitioning works is key to leveraging Kafka effectively.

### Purpose of Partitioning

1. **Scalability**: A topic can handle more data than can fit on a single broker by distributing its partitions across multiple brokers.
2. **Parallelism**: Multiple consumers within a group can process different partitions concurrently, increasing overall consumption throughput.
3. **Ordering**: Partitions provide the unit of ordering in Kafka—messages within a partition are strictly ordered.

### Partition Keys and Partitioning Strategies

When a producer sends a message, it can specify a **partition key**. Kafka uses this key to determine which partition the message should be sent to. The default partitioning strategy typically involves hashing the key and taking the result modulo the number of partitions:

```
partition_index = hash(key) % num_partitions
```

This ensures that **all messages with the same key are always sent to the same partition**. This property is fundamental for maintaining order for related events.

**Built-in Partitioning Strategies** Modern Kafka clients offer several partitioning strategies:

1. **Default Partitioner:**
  - If a partition is explicitly specified, use it
  - If a key is provided, hash the key to determine the partition
  - If no key is provided, use a sticky partitioning strategy (older clients used round-robin)
2. **Sticky Partitioner** (default for keyless messages in newer clients):
  - Assigns batches of messages to the same partition until the batch is full or `linger.ms` is reached
  - Then switches to another partition
  - Improves batching efficiency compared to round-robin
3. **Round-Robin Partitioner:**
  - Distributes messages evenly across all partitions in a circular manner
  - Maximizes load balancing but may reduce batching efficiency
4. **Custom Partitioners:**
  - Implement custom logic for special use cases (e.g., geographic partitioning)

**Choosing a Partition Key** The choice of partition key is critical and depends on the ordering requirements of your application:

- **Order ID:** If all events related to a specific order must be processed sequentially, use the order ID as the key.
- **Customer ID:** If events related to a specific customer need ordering (e.g., for session tracking), use the customer ID.
- **Device ID:** For IoT scenarios, use the device ID to ensure events from the same device are processed in order.
- **Null Key:** If no specific ordering is required, sending messages with a null key causes the producer to distribute them using the sticky or round-robin strategy.

**Impact of Key Choice:**

- **Ordering:** Guarantees sequential processing for messages with the same key.
- **Distribution:** A good key choice leads to even distribution across partitions. Poor key choices (e.g., keys with low cardinality or skewed distribution) can lead to “hot partitions” where some partitions receive significantly more traffic than others, creating bottlenecks.

```
// Example: Publishing with a partition key in Go
err = producer.Produce(&kafka.Message{
    TopicPartition: kafka.TopicPartition{
        Topic:      &topic,
        Partition: kafka.PartitionAny, // Let Kafka determine partition based on key
    },
```

```

    Key:    []byte(event.OrderID),    // Use OrderID as the key
    Value:  payload,
}, nil)

```

## Partition Skew and Hot Partitions

Partition skew occurs when some partitions receive significantly more data or requests than others. This can happen due to:

1. **Uneven Key Distribution:** If the partition key has a skewed distribution (e.g., some values are much more common than others).
2. **Poor Hashing:** If the hash function doesn't distribute keys evenly.
3. **Changing Partition Count:** When partitions are added, the mapping of keys to partitions changes, potentially causing temporary skew.

Hot partitions can cause several problems:

- **Throughput Bottlenecks:** The overall throughput is limited by the most heavily loaded partition.
- **Uneven Broker Load:** Brokers hosting hot partitions experience higher resource utilization.
- **Consumer Lag:** Consumers processing hot partitions may fall behind.

### Mitigating Partition Skew:

- **Choose High-Cardinality Keys:** Use keys with many possible values that are evenly distributed.
- **Composite Keys:** Combine multiple fields to create a more evenly distributed key.
- **Custom Partitioning:** Implement a custom partitioner that accounts for known skew in your data.
- **Monitor Partition Metrics:** Track bytes-in and message counts per partition to identify skew early.

## Ordering Guarantees Revisited

It's crucial to reiterate Kafka's ordering guarantees:

- **Within a partition:** Messages are strictly ordered based on their offset.
- **Across partitions:** There is **no** guaranteed order between messages in different partitions.

If your application requires global ordering across all events in a topic, you must use a topic with only a single partition. However, this severely limits scalability and parallelism and is generally discouraged unless absolutely necessary.

As noted in “Kafka in Action,” “Understanding and leveraging Kafka's ordering guarantees is essential for designing correct event-driven systems. Many distributed systems problems can be solved by ensuring related events are processed in order.”

## Consumer Groups and Rebalancing Dynamics

Consumer groups are Kafka's mechanism for enabling parallel consumption while ensuring that each message is processed by only one consumer within the group.

### Managing Offsets

Each consumer group maintains its own set of committed offsets for each partition it consumes. The committed offset represents the position up to which the group has successfully processed messages.

**Offset Storage** Offsets are stored in a special Kafka topic called `__consumer_offsets`. Each consumer group periodically commits its current position in each partition to this topic. The key format is:

```
[group_id, topic, partition]
```

And the value contains:

```
[offset, metadata, timestamp]
```

**Offset Commit Strategies** Consumers can commit offsets in several ways:

1. **Auto-commit** (`enable.auto.commit=true, auto.commit.interval.ms`):
  - The consumer automatically commits offsets periodically in the background
  - Simple but can lead to:
    - **At-most-once delivery**: If a crash occurs after processing but before the next auto-commit
    - **Duplicate processing**: If a crash occurs after auto-commit but before processing completes
2. **Synchronous Manual Commit** (`commitSync()`):
  - The application explicitly commits offsets and waits for acknowledgment
  - Provides better control but blocks until the commit succeeds
  - Typically used after processing a batch of records
3. **Asynchronous Manual Commit** (`commitAsync()`):
  - The application explicitly commits offsets without waiting for acknowledgment
  - Non-blocking but provides no guarantee that the commit succeeded
  - Often used with a callback to handle commit failures
4. **Exactly-Once Commit** (Transactions API):
  - Atomically commits both processing results and offsets
  - Requires transactional producers and consumers

```
// Example: Manual offset commit in Kotlin with Spring Kafka
@KafkaListener(topics = ["order.created"], groupId = "payment-service")
fun processOrder(record: ConsumerRecord<String, ByteArray>, acknowledgment: Acknowledgment) {
    try {
        // Deserialize and process the event
        val orderCreated = OrderCreatedEvent.parseFrom(record.value())
        processPayment(orderCreated)
    }
}
```

```

        // Manually acknowledge (commit) the offset after successful processing
        acknowledged
(Content truncated due to size limit. Use line ranges to read in chunks)

```

## # Chapter 6: Delivery Guarantees in Practice (Enhanced)

In distributed systems, particularly those built around event-driven architectures, ensuring

### ## The Spectrum of Delivery Guarantees

Kafka offers a spectrum of delivery guarantees, each with its own trade-offs between perform

#### ### At Most Once: Prioritizing Performance

At the performance-focused end of the spectrum, "at most once" delivery ensures that messa

In practice, "at most once" semantics mean that messages may be lost in various failure sc

- If a broker fails before replicating a message
- If a consumer crashes after reading a message but before processing it
- If a network partition occurs during message transmission

#### \*\*Configuration for At Most Once:\*\*

- **\*\*Producer\*\*:**
  - `acks=0` (don't wait for any acknowledgments)
  - `retries=0` (don't retry failed sends)
- **\*\*Consumer\*\*:**
  - `enable.auto.commit=true` (auto-commit offsets)
  - `auto.commit.interval.ms=100` (commit frequently)
  - Process messages after fetching but before the next commit interval

```
```go
```

```
// Example: At Most Once producer configuration in Go
```

```
producer, err := kafka.NewProducer(&kafka.ConfigMap{
    "bootstrap.servers": "kafka:9092",
    "acks":               "0", // Don't wait for any acknowledgments
    "retries":            0,   // Don't retry failed sends
})
```

```
// Example: At Most Once consumer configuration in Go
```

```
consumer, err := kafka.NewConsumer(&kafka.ConfigMap{
    "bootstrap.servers": "kafka:9092",
    "group.id":          "my-group",
    "auto.offset.reset": "latest",
    "enable.auto.commit": true,
    "auto.commit.interval.ms": 100, // Commit offsets every 100ms
})
```

As Scott, Gamov, and Klein note in “Kafka in Action,” at-most-once delivery “is suitable for use cases where occasional data loss is acceptable, such as metrics

collection or real-time analytics where approximate results are sufficient.”

### At Least Once: The Default Approach

In the middle of the spectrum, “at least once” delivery ensures that messages are delivered one or more times. This is Kafka’s default behavior and strikes a balance between reliability and complexity.

With “at least once” semantics, messages are guaranteed to be delivered, but they may be delivered multiple times in failure scenarios: - If a producer doesn’t receive an acknowledgment (even though the message was successfully stored) - If a consumer crashes after processing a message but before committing its offset - If a consumer rebalance occurs and offsets aren’t committed properly

**Configuration for At Least Once:** - **Producer:** - `acks=all` (wait for all in-sync replicas to acknowledge) - `retries=Integer.MAX_VALUE` (retry indefinitely) - `max.in.flight.requests.per.connection=1` (ensure ordering during retries) - `delivery.timeout.ms` (set appropriately for your use case) - **Consumer:** - `enable.auto.commit=false` (disable auto-commit) - Manually commit offsets after processing messages

```
// Example: At Least Once producer configuration in Kotlin
val producerProps = mapOf(
    ProducerConfig.BOOTSTRAP_SERVERS_CONFIG to "kafka:9092",
    ProducerConfig.ACKS_CONFIG to "all", // Wait for all in-sync replicas
    ProducerConfig.RETRIES_CONFIG to Int.MAX_VALUE, // Retry indefinitely
    ProducerConfig.MAX_IN_FLIGHT_REQUESTS_PER_CONNECTION to 1, // Ensure ordering during
    ProducerConfig.DELIVERY_TIMEOUT_MS_CONFIG to 120000 // 2 minutes max delivery time
)

// Example: At Least Once consumer in Kotlin with Spring Kafka
@KafkaListener(topics = ["my-topic"], groupId = "my-group")
fun processMessage(
    record: ConsumerRecord<String, String>,
    acknowledgment: Acknowledgment
) {
    try {
        // Process the message
        processMessage(record.value())

        // Commit offset only after successful processing
        acknowledgment.acknowledge()
    } catch (e: Exception) {
        // Handle error, but don't commit offset
        // The message will be redelivered
        logger.error("Error processing message: ${e.message}")
    }
}
```

Bejeck emphasizes in “Kafka Streams in Action” that “at-least-once delivery is the most common approach in practice, as it balances reliability with implementation complexity. When combined with idempotent consumers, it provides a



robust solution for most use cases.”

### Exactly Once Semantics (EOS): The Holy Grail

At the reliability-focused end of the spectrum, “exactly once” semantics ensure that messages are delivered exactly once, even in the presence of failures. This is the most complex guarantee to implement but provides the strongest reliability.

Kafka supports exactly once semantics through its transactions API, which was introduced in version 0.11.0 and enhanced in subsequent releases. However, it’s important to note that true exactly once processing is only guaranteed within the Kafka ecosystem—when integrating with external systems, additional coordination mechanisms are required.

**Configuration for Exactly Once:**

- **Producer:** - `enable.idempotence=true` (prevent duplicates from producer retries) - `transactional.id=<unique-id>` (enable transactions) - `acks=all` (implied by `enable.idempotence`) - `retries=Integer.MAX_VALUE` (implied by `enable.idempotence`)
- **Consumer:** - `isolation.level=read_committed` (only read committed messages) - `enable.auto.commit=false` (manual offset management)

```
// Example: Exactly Once producer configuration in Java
Properties props = new Properties();
props.put(ProducerConfig.BOOTSTRAP_SERVERS_CONFIG, "kafka:9092");
props.put(ProducerConfig.ENABLE_IDEMPOTENCE_CONFIG, true);
props.put(ProducerConfig.TRANSACTIONAL_ID_CONFIG, "my-transactional-id");

KafkaProducer<String, String> producer = new KafkaProducer<>(props);
producer.initTransactions();

// Example: Exactly Once processing (read-process-write pattern)
producer.beginTransaction();
try {
    // Consume, process, and produce in a single transaction
    ConsumerRecords<String, String> records = consumer.poll(Duration.ofMillis(100));
    for (ConsumerRecord<String, String> record : records) {
        // Process the record
        String result = processRecord(record.value());

        // Produce the result
        producer.send(new ProducerRecord<>("output-topic", result));
    }

    // Commit consumer offsets and producer messages atomically
    producer.sendOffsetsToTransaction(
        getOffsets(records),
        consumer.groupMetadata()
    );

    // Commit the transaction
    producer.commitTransaction();
}
```

```

} catch (Exception e) {
    // Abort the transaction on error
    producer.abortTransaction();
    throw e;
}

```

## Understanding Kafka's Idempotent Producer

Introduced in Kafka 0.11, the idempotent producer is a key component of Kafka's exactly-once semantics. When enabled (`enable.idempotence=true`), the producer assigns each batch of messages a sequence number and the broker tracks these sequence numbers to prevent duplicate writes.

The idempotent producer works by:

1. Assigning a Producer ID (PID) to each producer instance
2. Maintaining sequence numbers for each producer-partition pair
3. Rejecting messages with outdated sequence numbers
4. Acknowledging messages with duplicate sequence numbers without writing them again

This mechanism prevents duplicates caused by producer retries, but it has limitations:

- It only works within a single producer session (if the producer restarts, it gets a new PID)
- It doesn't prevent duplicates from application-level retries (e.g., if your application retries a failed send with a new producer)

To overcome these limitations, the transactional producer extends the idempotent producer with a persistent `transactional.id` that survives restarts.

## The Myth of Exactly Once

While "exactly once" semantics are technically achievable within Kafka, they come with significant caveats and limitations. It's important to understand these nuances to make informed architectural decisions.

### Limitations of Kafka's Exactly Once

1. **Scope:** Exactly once semantics only apply within the Kafka ecosystem. When integrating with external systems (databases, APIs, etc.), additional coordination mechanisms are required.
2. **Performance Impact:** Transactions add overhead in terms of latency and throughput. According to benchmarks cited in "Kafka in Action," transactional producers can be 20-40% slower than regular producers, depending on the workload.
3. **Complexity:** Implementing exactly once semantics correctly requires careful configuration and error handling. Mistakes in implementation can lead to unexpected behavior.
4. **Failure Modes:** Even with transactions, certain failure scenarios can still lead to message duplication or loss:
  - If the transaction coordinator fails during a transaction
  - If a consumer crashes after processing but before the transaction commits
  - If network partitions last longer than the transaction timeout

5. **Operational Overhead:** Transactions require additional monitoring and management, including:

- Transaction log cleanup (`transactional.id.expiration.ms`)
- Transaction timeout configuration (`transaction.timeout.ms`)
- Monitoring transaction failures and aborts

As noted by Gamov in “Kafka in Action,” “Exactly-once semantics are powerful but come with a cost. For many applications, at-least-once delivery with idempotent consumers provides a better balance of reliability and performance.”

### The Pragmatic Alternative: Idempotent Processing

Given these limitations, many practitioners opt for a more pragmatic approach: “at least once” delivery combined with idempotent processing. This approach acknowledges that duplicates may occur but ensures they don’t affect the system’s correctness.

**Idempotent processing** means that applying the same operation multiple times has the same effect as applying it once. In the context of message processing, it means that processing the same message multiple times doesn’t change the system’s state beyond the first processing.

### Strategies for Implementing Idempotent Consumers

1. **Natural Idempotence:** Some operations are naturally idempotent:
  - Setting a value (e.g., `user.status = "active"`)
  - Conditional updates (e.g., `if user.status != "active" then user.status = "active"`)
  - Absolute operations (e.g., `inventory.quantity = 100` vs. `inventory.quantity += 10`)
2. **Deduplication by Message ID:** Track processed message IDs to detect and skip duplicates:

```
# Example: Idempotent message processing in Ruby
def process_payment(payment_event)
  # Extract a unique identifier from the event
  payment_id = payment_event.payment_id

  # Check if we've already processed this payment
  if redis.exists?("processed:payment:#{payment_id}")
    logger.info("Payment #{payment_id} already processed, skipping")
    return
  end

  # Process the payment
  result = payment_gateway.process(
    amount: payment_event.amount,
    customer_id: payment_event.customer_id,
    payment_method: payment_event.payment_method
  )
end
```

```

# Record the result
db.transaction do
  # Store the payment result
  Payment.create!(
    id: payment_id,
    order_id: payment_event.order_id,
    amount: payment_event.amount,
    status: result.success? ? 'completed' : 'failed',
    transaction_id: result.transaction_id
  )

  # Mark as processed with a reasonable TTL
  redis.set("processed:payment:#{payment_id}", "1", ex: 7.days.to_i)
end

# Publish a result event
publish_payment_processed_event(payment_id, result)
end

```

3. **Event Versioning:** Include a version or timestamp in events and only process newer versions:

```

// Example: Event versioning for idempotence
data class OrderUpdatedEvent(
  val orderId: String,
  val version: Long, // Monotonically increasing version number
  val status: String,
  val updatedAt: Instant
)

fun processOrderUpdate(event: OrderUpdatedEvent) {
  // Fetch current order from database
  val order = orderRepository.findById(event.orderId)
    ?: throw OrderNotFoundException(event.orderId)

  // Only apply the update if the event version is newer
  if (event.version > order.version) {
    order.status = event.status
    order.version = event.version
    order.updatedAt = event.updatedAt
    orderRepository.save(order)
  } else {
    logger.info("Skipping outdated event for order ${event.orderId}: event version ${e
  }
}

```

4. **Idempotent APIs:** Design downstream APIs to be idempotent using request IDs:

```

// Example: Idempotent API call
func processShipment(shipmentEvent *ShipmentEvent) error {
  // Create idempotent request with unique ID

```

```

req := &ShipmentRequest{
    IdempotencyKey: shipmentEvent.EventID,
    OrderID:       shipmentEvent.OrderID,
    Address:       shipmentEvent.ShippingAddress,
    Items:         shipmentEvent.Items,
}

// The shipping API will use the idempotency key to deduplicate requests
resp, err := shippingClient.CreateShipment(context.Background(), req)
if err != nil {
    if isIdempotencyConflict(err) {
        // This is a duplicate request, the shipment was already created
        log.Printf("Duplicate shipment request detected: %s", shipmentEvent.EventID)
        return nil
    }
    return err
}

return nil
}

```

This approach is often simpler to implement and reason about than true exactly once processing, while still providing the necessary correctness guarantees for most applications.

## Verifying Delivery Guarantees Through Testing

Theoretical guarantees are valuable, but nothing beats empirical verification through testing. Let's explore practical approaches to testing Kafka's delivery guarantees.

### Chaos Testing: Breaking Things on Purpose

Chaos testing involves deliberately introducing failures into your system to observe how it behaves. For Kafka delivery guarantees, this means simulating various failure scenarios:

1. **Broker Failures:** Killing Kafka brokers during message production and consumption
2. **Network Partitions:** Introducing network delays or disconnections between clients and brokers
3. **Consumer Crashes:** Forcibly terminating consumers during message processing
4. **Producer Crashes:** Killing producers before they receive acknowledgments

Here's a simplified example of a chaos test script that kills a consumer midway through processing:

```

#!/bin/bash
# chaos_test.sh - Test consumer resilience to crashes

```

```

# Start the consumer in the background
./start_consumer.sh &
CONSUMER_PID=$!

# Wait for the consumer to start processing
sleep 5

# Send a batch of test messages with unique IDs
for i in {1..1000}; do
    ./send_test_message.sh "test-message-$i"
done

# Wait for some messages to be processed
sleep 10

# Kill the consumer abruptly
kill -9 $CONSUMER_PID

# Wait a moment
sleep 2

# Restart the consumer
./start_consumer.sh &
NEW_CONSUMER_PID=$!

# Wait for processing to complete
sleep 30

# Check the results
./verify_processed_m
(Content truncated due to size limit. Use line ranges to read in chunks)

# Chapter 7: Go: Efficient Simplicity with Watermill

```

Welcome to Part III of our journey, where we transition from the foundational principles of

As discussed in Chapter 3, Go is often chosen for microservices due to its performance cha

*## Introducing Watermill: An Idiomatic Go Library for Event-Driven Apps*

Watermill is a Go library designed to simplify the development of event-driven application

Key features of Watermill relevant to our Kafka implementation include:

- **\*\*Publisher/Subscriber Abstractions\*\***: Provides unified interfaces for publishing and su
- **\*\*Router\*\***: A powerful component for defining message handlers, middleware, and processi
- **\*\*Middleware\*\***: Offers built-in middleware for common tasks like retries, throttling, co
- **\*\*Pluggable Architecture\*\***: Supports various message brokers through specific implementa
- **\*\*Structured Logging\*\***: Integrates well with standard Go logging practices.

By using Watermill, we can focus more on our application logic and less on the low-level details.

*## Setting Up Watermill with Kafka in Go*

Let's examine how we configure Watermill to interact with Kafka in our Go services, drawing on the following example.

### Dependencies

First, we need to include the necessary Watermill packages in our `go.mod` file:

```
``go
// go.mod (simplified)
require (
    github.com/ThreeDotsLabs/watermill v1.3.5
    github.com/ThreeDotsLabs/watermill-kafka/v2 v2.5.0
    // ... other dependencies
)
```

We specifically need `watermill` for the core library and `watermill-kafka/v2` for the Kafka-specific Pub/Sub implementation.

### Creating a Kafka Publisher

To publish messages to Kafka, we create a Watermill Publisher. The `kafka.NewPublisher` function takes a configuration and a marshaler (for serializing messages).

```
// From kafka_content/go/pkg/watermillx/publisher.go (simplified)
import (
    "github.com/ThreeDotsLabs/watermill"
    "github.com/ThreeDotsLabs/watermill-kafka/v2/pkg/kafka"
    "github.com/ThreeDotsLabs/watermill/message"
)
```

## # Chapter 8: Kotlin: JVM Power with Spring Kafka

Continuing our exploration of polyglot Kafka integration in Part III, we now turn our attention to Spring Kafka.

Spring Kafka, part of the larger Spring ecosystem, provides high-level abstractions over the underlying Kafka API.

In this chapter, we'll examine how our reference implementation leverages Kotlin and Spring Kafka.

### ## Why Kotlin and Spring Kafka?

While Go (Chapter 7) excels in simplicity and raw performance, Kotlin on the JVM offers distinct advantages:

- **Rich Ecosystem**: Access to the extensive Java ecosystem, including mature libraries and frameworks.
- **Developer Productivity**: Kotlin's concise syntax and features like data classes, extension functions, and coroutines.
- **Spring Integration**: Spring Boot and Spring Kafka provide a highly opinionated yet flexible framework for building applications.
- **Maturity and Stability**: The JVM and Spring Framework are battle-tested platforms used by millions of developers.

Spring Kafka, in particular, simplifies Kafka interactions by providing:

- **Annotation-driven listeners**: Easy creation of consumers using the `@KafkaListener`.
- **Template-based producers**: Simplified message publishing via `KafkaTemplate`.
- **Error handling and retries**: Built-in support for handling exceptions and retrying failed messages.
- **Transaction management**: Integration with Spring's transaction abstraction for exactly-once semantics.
- **Serialization/Deserialization**: Flexible configuration for message converters (e.g., `StringSerializer`, `StringDeserializer`, `ByteArraySerializer`, `ByteArrayDeserializer`).

## ## Setting Up Spring Kafka in Kotlin

Let's look at the core configuration required to integrate Spring Kafka into our Kotlin-based application.

### ### Dependencies

We need to include the necessary Spring Boot and Spring Kafka dependencies in our `build.gradle.kts`:

```
```kotlin
// build.gradle.kts (simplified)
dependencies {
    implementation("org.springframework.boot:spring-boot-starter")
    implementation("org.springframework.kafka:spring-kafka")
    implementation("org.jetbrains.kotlin:kotlin-reflect")
    implementation("org.jetbrains.kotlin:kotlin-stdlib-jdk8")
    implementation("com.google.protobuf:protobuf-java:3.19.4") // For Protobuf
    // ... other dependencies like Spring Web, Spring Data JPA, etc.
}
```

### Configuration (application.yml)

Spring Boot allows configuring Kafka connection details and other properties in `application.yml` or `application.properties`:

```
# application.yml
spring:
  application:
    name: shipping-service
  kafka:
    bootstrap-servers: kafka:9092 # Comma-separated list of brokers
    consumer:
      group-id: shipping-service-group # Default consumer group ID
      auto-offset-reset: earliest # Start consuming from the beginning if no offset exists
      key-deserializer: org.apache.kafka.common.serialization.StringDeserializer
      value-deserializer: org.apache.kafka.common.serialization.ByteArrayDeserializer # Using Protobuf
    producer:
      key-serializer: org.apache.kafka.common.serialization.StringSerializer
      value-serializer: org.apache.kafka.common.serialization.ByteArraySerializer # Using Protobuf
      acks: all # Ensure producer waits for all replicas
```

This configuration sets up the broker addresses, default consumer group ID, and serializers/deserializers. We use `ByteArrayDeserializer` and `ByteArraySerializer` because we are handling Protobuf serialization manually.



ally in our service code.

### Kafka Configuration Beans (KafkaConfig.kt)

While `application.yml` provides basic settings, we often define explicit configuration beans for more control, especially over factories and templates. Our reference implementation does this in `KafkaConfig.kt`:

```
// From kafka_content/kotlin-service/src/main/kotlin/com/scrapybara/kw/shipping/ShippingService.kt
@Configuration
class KafkaConfig {
    @Value("\${spring.kafka.bootstrap-servers}")
    private lateinit var bootstrapServers: String

    // Producer Factory: Creates KafkaProducer instances
    @Bean
    fun producerFactory(): ProducerFactory<String, ByteArray> {
        val configProps = mapOf(
            ProducerConfig.BOOTSTRAP_SERVERS_CONFIG to bootstrapServers,
            ProducerConfig.KEY_SERIALIZER_CLASS_CONFIG to StringSerializer::class.java,
            ProducerConfig.VALUE_SERIALIZER_CLASS_CONFIG to ByteArraySerializer::class.java,
            ProducerConfig.ACKS_CONFIG to "all" // Moved from application.yml for clarity
        )
        return DefaultKafkaProducerFactory(configProps)
    }

    // KafkaTemplate: High-level abstraction for sending messages
    @Bean
    fun kafkaTemplate(producerFactory: ProducerFactory<String, ByteArray>): KafkaTemplate<String, ByteArray> {
        return KafkaTemplate(producerFactory)
    }

    // Consumer Factory: Creates KafkaConsumer instances
    @Bean
    fun consumerFactory(): ConsumerFactory<String, ByteArray> {
        val props = mapOf(
            ConsumerConfig.BOOTSTRAP_SERVERS_CONFIG to bootstrapServers,
            ConsumerConfig.GROUP_ID_CONFIG to "shipping-service-group", // Explicit group id
            ConsumerConfig.KEY_DESERIALIZER_CLASS_CONFIG to StringDeserializer::class.java,
            ConsumerConfig.VALUE_DESERIALIZER_CLASS_CONFIG to ByteArrayDeserializer::class.java,
            ConsumerConfig.AUTO_OFFSET_RESET_CONFIG to "earliest"
        )
        return DefaultKafkaConsumerFactory(props)
    }

    // Listener Container Factory: Creates containers for @KafkaListener methods
    @Bean
    fun kafkaListenerContainerFactory(consumerFactory: ConsumerFactory<String, ByteArray>): KafkaListenerContainerFactory<String, ByteArray> {
        val factory = ConcurrentKafkaListenerContainerFactory<String, ByteArray>()
        factory.consumerFactory = consumerFactory
    }
}
```

```

        // Additional configuration (e.g., concurrency, error handling) can go here
        return factory
    }
}

```

This code defines beans for `ProducerFactory`, `KafkaTemplate`, `ConsumerFactory`, and `ConcurrentKafkaListenerContainerFactory`. These factories are used by Spring Kafka under the hood to create producers and consumers. The `KafkaTemplate` is what we inject into our services to publish messages.

## Consuming Messages with `@KafkaListener`

Spring Kafka makes consuming messages incredibly simple using the `@KafkaListener` annotation on a method within a Spring-managed bean (e.g., a `@Service`).

```

// From kafka_content/kotlin-service/src/main/kotlin/com/scrapybara/kw/shipping/ShippingService.kt
@Service
class ShippingService(
    // ... other dependencies
    private val shippingSagaManager: ShippingSagaManager,
    private val metrics: ShippingMetrics,
    private val registry: MeterRegistry
) {
    private val logger = LoggerFactory.getLogger(ShippingService::class.java)

    @KafkaListener(topics = ["inventory.checked"], containerFactory = "kafkaListenerContainerFactory")
    fun handleInventoryChecked(payloadBytes: ByteArray) {
        metrics.kafkaMessagesReceived.increment()
        val sample = Timer.start(registry)
        try {
            // 1. Deserialize using Protobuf
            val event = OrderProto.InventoryChecked.parseFrom(payloadBytes)
            logger.info("Received inventory checked event for order: ${event.orderId}, allItemsAvailable: ${event.allItemsAvailable}")

            // 2. Delegate to business logic (Saga Manager in this case)
            // Using runBlocking here for simplicity in the example, consider structured concurrency in production
            runBlocking {
                shippingSagaManager.startShippingSaga(
                    orderId = event.orderId,
                    allItemsAvailable = event.allItemsAvailable
                )
            }

            // 3. Update metrics based on outcome
            if (event.allItemsAvailable) {
                metrics.incrementOrdersFulfilled()
            } else {
                metrics.incrementOrdersCancelled()
            }
        } finally {
            metrics.ordersProcessed.increment()
        }
    }
}

```

```

    } catch (e: Exception) {
        logger.error("Error processing inventory checked event", e)
        metrics.kafkaMessagesFailed.increment()
        // Re-throw to let Spring Kafka handle retries/error handling
        throw e
    } finally {
        // Record processing time metric
        sample?.stop(metrics.shippingProcessingTime)
    }
}
// ... other listener methods for different topics ...
}

```

Key aspects of this listener:

- **@KafkaListener**: Marks the method as a Kafka message consumer.
- **topics**: Specifies the Kafka topic(s) to subscribe to.
- **containerFactory**: Refers to the bean name of the `ConcurrentKafkaListenerContainerFactory` we defined earlier. This factory configures the underlying consumer container.
- **Method Parameter (payloadBytes: ByteArray)**: Spring Kafka injects the raw message payload as a byte array (because we configured `ByteArrayDeserializer`).
- **Deserialization**: We manually deserialize the byte array using the generated Protobuf code (`OrderProto.InventoryChecked.parseFrom(payloadBytes)`).
- **Business Logic**: The core logic is delegated to other components (`shippingSagaManager`).
- **Error Handling**: Exceptions are caught, logged, and re-thrown. Spring Kafka's default error handling mechanisms (including potential retries) will then take over.
- **Acknowledgment**: By default, if the listener method completes without throwing an exception, Spring Kafka automatically acknowledges the message (commits the offset).

## Publishing Messages with `KafkaTemplate`

To publish messages, we inject the `KafkaTemplate` bean (defined in `KafkaConfig.kt`) into our service and use its `send` method.

```

// From kafka_content/kotlin-service/src/main/kotlin/com/scrapybara/kw/shipping/config/KafkaConfig.kt
@Component
class KafkaOperations(private val kafkaTemplate: KafkaTemplate<String, ByteArray>) {
    private val logger = LoggerFactory.getLogger(KafkaOperations::class.java)

    fun publishShipmentCreated(event: OrderProto.ShipmentCreated) {
        val topic = "shipment.created"
        val key = event.orderId // Use orderId as partition key
        val payload = event.toByteArray()

        logger.info("Publishing ShipmentCreated event for order: ${event.orderId} to topic")
    }
}

```

```

        kafkaTemplate.send(topic, key, payload)
            .addCallback(
                { result -> logger.info("Successfully published event to ${result?.recordMetadata?.topic}", result) },
                { ex -> logger.error("Failed to publish event for order ${event.orderId}", ex) }
            )
    }

    // ... other publish methods for different event types ...
}

// Usage within another service:
@Service
class ShippingSaga(/* ... */, private val kafkaOps: KafkaOperations) {
    suspend fun createShipment(orderId: String, /* ... */) {
        // ... logic to create shipment ...

        val shipmentCreatedEvent = OrderProto.ShipmentCreated.newBuilder()
            .setShipmentId(shipment.id)
            .setOrderId(orderId)
            // ... set other fields ...
            .build()

        kafkaOps.publishShipmentCreated(shipmentCreatedEvent)
    }
}

```

Key aspects of publishing:

- **KafkaTemplate Injection:** The template is injected via constructor injection.
- **send(topic, key, payload):** The primary method for sending messages. We provide the topic name, the partition key (crucial for ordering, using `orderId` here), and the serialized payload (byte array from Protobuf).
- **Serialization:** We manually serialize the Protobuf object to a byte array using `event.toByteArray()`.
- **Asynchronous Sending:** `kafkaTemplate.send` returns a `ListenableFuture` (or `CompletableFuture` in newer versions). We use `addCallback` to handle success and failure asynchronously, preventing the main thread from blocking.

## Error Handling and Retries in Spring Kafka

Spring Kafka provides several mechanisms for handling errors during message consumption.

### Default Behavior

If a `@KafkaListener` method throws an exception, Spring Kafka's default error handler logs the error but does not stop consumption. The offset is not committed, meaning the failed message (and subsequent messages in the same partition) will likely be redelivered on the next poll.

## Retries with Backoff

Spring Retry can be integrated to automatically retry failed listener executions with configurable backoff policies. This is often enabled via `@EnableRetry` and configuring a `RetryTemplate` or using `@Retryable` annotations (though listener retries are often configured at the container factory level).

Our reference implementation includes a `RetryConfig.kt` which likely configures this, although the specific mechanism (e.g., `SeekToCurrentErrorHandler` with backoff) isn't shown in the provided snippets.

```
// Example configuration snippet (conceptual)
@Bean
fun kafkaListenerContainerFactory(/* ... */): ConcurrentKafkaListenerContainerFactory<String> {
    // ... other factory config ...

    // Configure error handler with retries and backoff
    factory.setErrorHandler(SeekToCurrentErrorHandler(
        FixedBackOff(1000L, 3L) // 1-second interval, 3 retries
    ))

    return factory
}
```

## Dead Letter Queues (DLQ)

After exhausting retries, it's common practice to send the persistently failing message to a Dead Letter Queue (DLQ) topic for later inspection. Spring Kafka supports this through error handlers like `SeekToCurrentErrorHandler` or custom configurations.

```
// Example configuration snippet (conceptual)
@Bean
fun kafkaListenerContainerFactory(/* ... */): ConcurrentKafkaListenerContainerFactory<String> {
    // ... other factory config ...

    // Configure error handler with DLQ
    val recoverer = DeadLetterPublishingRecoverer(kafkaTemplate) // Pass KafkaTemplate
    factory.setErrorHandler(SeekToCurrentErrorHandler(
        recoverer,
        FixedBackOff(1000L, 3L)
    ))

    return factory
}
```

The `DeadLetterPublishingRecoverer` automatically sends the failed message to a topic named `<original-topic-name>.DLT` by default, adding headers with error information.

## Integrating with Kotlin Coroutines

Kotlin's coroutines provide excellent support for asynchronous programming. While Spring Kafka's listener model is inherently synchronous per thread, coroutines can be used within the listener method for non-blocking I/O operations or complex asynchronous logic.

```
@KafkaListener(topics = ["some.topic"])
fun handleEvent(payloadBytes: ByteArray) = runBlocking { // Use runBlocking or la
(Content truncated due to size limit. Use line ranges to read in chunks)
```

# Chapter 9: Ruby: Dynamic Productivity with ruby-kafka

Completing our tour of the polyglot landscape in Part III, we arrive at Ruby—a language ce

In our reference implementation, Ruby powers the Analytics service (``kafka_content/ruby-se`

## The Role of Ruby in Our System

While Go handles high-throughput core services and Kotlin manages complex stateful process

- **Expressiveness**: Ruby code is often concise and reads almost like natural language, m
- **Data Manipulation**: Ruby's rich set of built-in methods for arrays, hashes, and strin
- **Rapid Development**: The dynamic nature of Ruby allows for quick iteration and adaptat
- **Ecosystem**: Gems for data processing, web serving (like Sinatra, used here), and data

## Introducing `ruby-kafka`

`ruby-kafka` is the most widely used and actively maintained Kafka client library for Ruby

Key features of `ruby-kafka` include:

- **Producer API**: Methods for sending messages synchronously or asynchronously, with sup
- **Consumer API**: Support for consumer groups, manual and automatic offset management, a
- **Broker Discovery**: Automatic discovery of brokers in the cluster.
- **Compression**: Support for Gzip and Snappy compression.
- **SSL/SASL Authentication**: Secure communication with Kafka brokers.

## Setting Up `ruby-kafka`

Integrating `ruby-kafka` into our Ruby service involves adding the gem and configuring the

### Dependencies (`Gemfile`)

First, we add `ruby-kafka` to our project's `Gemfile`:

```
``ruby
# Gemfile
source "https://rubygems.org"
```

```
gem "kafka", "~> 1.4" # Or the latest version of ruby-kafka
gem "sinatra", "~> 3.0" # For the web interface
gem "protobuf", "~> 3.19" # For Protobuf serialization
gem "concurrent-ruby" # For background processing
# ... other gems
```

Then, run `bundle install` to install the dependencies.

## Configuring the Kafka Client

We initialize the `ruby-kafka` client, providing the necessary configuration, such as the list of seed brokers and a client ID.

```
# From kafka_content/ruby-service/analytics_service.rb
require 'kafka'
require 'logger'

# Set up logging
logger = Logger.new(STDOUT)
logger.level = Logger::INFO

# Configure Kafka client
kafka = Kafka.new(
  seed_brokers: ['kafka:9092'], # List of initial brokers to connect to
  client_id: 'analytics-service', # Identifier for this client instance
  logger: logger # Integrate with Ruby's standard logger
)
```

`ruby-kafka` uses the `seed_brokers` to connect to the cluster and discover the rest of the brokers. The `client_id` helps identify this specific client instance in Kafka logs and metrics.

## Consuming Messages with `ruby-kafka`

The Analytics service primarily acts as a consumer, listening to various topics. `ruby-kafka` provides a straightforward way to create consumers and subscribe to topics.

### Creating a Consumer

We create a consumer instance associated with a specific `group_id`. This ID is essential for Kafka's consumer group functionality, ensuring that partitions are distributed among consumers in the group and offsets are tracked correctly.

```
# From kafka_content/ruby-service/analytics_service.rb
consumer = kafka.consumer(
  group_id: 'analytics-service-group' # Unique ID for this consumer group
)
```

### Subscribing to Topics

Once the consumer is created, we subscribe it to the topics we're interested in. The Analytics service subscribes to multiple topics related to the order lifecycle.

```

# From kafka_content/ruby-service/analytics_service.rb
consumer.subscribe('order.created')
consumer.subscribe('payment.processed')
consumer.subscribe('inventory.checked') # Assuming this topic exists
consumer.subscribe('order.fulfilled')
consumer.subscribe('order.cancelled')

```

## Processing Messages

The `each_message` method provides a simple blocking loop that yields each message received from the subscribed topics. This is the core of our consumer logic.

```

# From kafka_content/ruby-service/analytics_service.rb

# Initialize analytics data structure (simple example)
order_counts = {
  created: 0,
  payment_success: 0,
  payment_failed: 0,
  fulfilled: 0,
  cancelled: 0
}

begin
  consumer.each_message do |message|
    topic = message.topic
    value = message.value # Raw message payload (bytes)
    key = message.key      # Partition key (bytes or nil)
    offset = message.offset
    partition = message.partition

    logger.debug("Received message from topic=#{topic}, partition=#{partition}, offset=#{offset}")

    begin
      # 1. Deserialize based on topic (assuming Protobuf)
      case topic
      when 'order.created'
        # Assumes Protobuf generated classes exist in KafkaWatermill::IDL
        event = KafkaWatermill::IDL::OrderCreated.decode(value)
        order_counts[:created] += 1
        logger.info("Order created event processed: #{event.order_id}")

      when 'payment.processed'
        event = KafkaWatermill::IDL::PaymentProcessed.decode(value)
        if event.status == 'COMPLETED' # Assuming status field exists
          order_counts[:payment_success] += 1
          logger.info("Payment success event processed: #{event.order_id}")
        else
          order_counts[:payment_failed] += 1
        end
      end
    end
  end
end

```



```

        logger.info("Payment failed event processed: #{event.order_id}")
    end

    when 'order.fulfilled'
        event = KafkaWatermill::IDL::OrderFulfilled.decode(value)
        order_counts[:fulfilled] += 1
        logger.info("Order fulfilled event processed: #{event.order_id}")

    when 'order.cancelled'
        event = KafkaWatermill::IDL::OrderCancelled.decode(value)
        order_counts[:cancelled] += 1
        logger.info("Order cancelled event processed: #{event.order_id}")

    # Add cases for other subscribed topics like 'inventory.checked'
    # when 'inventory.checked'
    #   event = KafkaWatermill::IDL::InventoryChecked.decode(value)
    #   logger.info("Inventory checked event processed: #{event.order_id}")

    else
        logger.warn("Received message from unexpected topic: #{topic}")
    end

    # 2. Update analytics (in this case, simple counts)
    # (Already done within the case statement)

    # 3. Log aggregated metrics periodically (example)
    if (order_counts[:created] % 10) == 0 && order_counts[:created] > 0
        logger.info("ANALYTICS REPORT: #{order_counts.inspect}")
    end

    # 4. Acknowledge message (handled automatically by each_message by default)

rescue => e
    # Handle deserialization or processing errors for a single message
    logger.error("Error processing message from topic #{topic} at offset #{offset}: #{e}")
    logger.error(e.backtrace.join("\n"))
    # Decide whether to stop the consumer or skip the message
    # By default, `each_message` continues on error
end

end

rescue => e
    # Handle consumer-level errors (e.g., connection issues)
    logger.error("Kafka consumer error: #{e.message}")
    logger.error(e.backtrace.join("\n"))
    # Consider exiting or attempting to reconnect
end
end

```

Key points in this processing loop:

- **each\_message Loop:** Continuously fetches and yields messages.
- **Deserialization:** The raw `message.value` needs to be deserialized. We

use a `case` statement based on the `message.topic` and assume corresponding Protobuf classes (`KafkaWatermill::IDL::*`) generated from our `.proto` files are available. The `decode` method is used for Protobuf deserialization.

- **Error Handling:** Includes `begin/rescue` blocks to catch errors during individual message processing and errors related to the consumer itself.
- **Offset Management:** By default, `each_message` automatically commits offsets after the block for a message executes successfully. This provides “at least once” delivery semantics. For more control, `each_message` can be configured with `automatically_mark_as_processed: false`, requiring manual offset marking.

## Integrating Protobuf with Ruby

Similar to Go and Kotlin, we rely on Protocol Buffers for schema definition and serialization across our polyglot services.

1. **Define Schemas:** Use the same `.proto` files as the other services.
2. **Generate Ruby Code:** Use `protoc` with the Ruby plugin (`gem install google-protobuf`) to generate Ruby classes (`*_pb.rb` files) corresponding to the Protobuf messages.

```
protoc --proto_path=./idl --ruby_out=lib/idl *.proto
```

3. **Require Generated Files:** Ensure the generated Ruby files are required in your application.

```
# Example: Assuming generated files are in lib/idl
require_relative 'lib/idl/order_events_pb'
# ... require other generated files ...
```

4. **Serialize/Deserialize:** Use the `encode` and `decode` methods provided by the generated classes.

```
# Deserialize
event = KafkaWatermill::IDL::OrderCreated.decode(message.value)
puts event.order_id

# Serialize (if producing)
# new_event = KafkaWatermill::IDL::AnalyticsUpdated.new(order_id: '123', metric: 'count',
# payload = new_event.encode
# producer.produce(payload, topic: 'analytics.updated', key: '123')
```

## Running the Consumer

The `each_message` loop is blocking. To allow other tasks (like running a web server) to execute concurrently, we run the consumer loop in a separate thread or use a concurrency library like `concurrent-ruby`.

```
# From kafka_content/ruby-service/analytics_service.rb
require 'concurrent'

# Start the analytics processor in a background thread using concurrent-ruby
```

```

processor = Concurrent::Promise.execute do
  logger.info('Analytics service starting to consume events...')
  # ... consumer.each_message loop as shown above ...
end

# ... Run Sinatra web server or other tasks ...

# Ensure graceful shutdown
at_exit do
  logger.info('Shutting down analytics service...')
  consumer.stop # Signals the consumer loop to terminate
end

# Optional: Wait for the processor promise if needed
# processor.wait!

```

The `at_exit` block is crucial for ensuring the consumer is stopped gracefully when the application terminates, allowing it to commit final offsets and leave the consumer group cleanly.

## Exposing Analytics via a Web Interface (Sinatra)

The reference implementation uses the lightweight Sinatra web framework to expose the aggregated analytics data via a simple HTTP endpoint.

```

# From kafka_content/ruby-service/analytics_service.rb
require 'sinatra'
require 'json'

class AnalyticsAPI < Sinatra::Base
  set :bind, '0.0.0.0' # Bind to all interfaces
  set :port, 3000      # Run on port 3000

  # Make analytics data accessible (needs proper scoping/sharing)
  # This assumes order_counts is accessible here; consider using a class
  # or other mechanism for sharing state between the consumer thread and web server.
  # For simplicity, let's assume it's globally accessible (not recommended for production)
  $analytics_data = order_counts

  get '/analytics' do
    content_type :json
    {
      order_counts: $analytics_data,
      timestamp: Time.now.iso8601
    }.to_json
  end

  get '/health' do
    content_type :json
    { status: 'UP' }.to_json
  end
end

```

end

```
# Start the Sinatra app (blocks the main thread)
AnalyticsAPI.run!
```

This demonstrates how easily other Ruby gems can be integrated alongside `ruby-kafka` to build a complete service.

## Considerations for Production Ruby Services

While the reference implementation provides a functional example, building production-grade Ruby Kafka consumers involves additional considerations:

- **Concurrency:** The `each_message` loop processes messages sequentially within a single thread. For higher throughput, you might need multiple consumer processes or explore libraries that offer thread-based parallel processing within a single consumer instance (though this can complicate offset management).
- **Error Handling:** Implement robust strategies for handling persistent message failures, potentially including manual DLQ logic if automatic retries are insufficient.
- **Offset Management:** For critical applications, consider manual offset commits (`automatically_mark_as_processed: false`) to ensure messages are only acknowledged after successful processing, potentially involving database transactions (e.g., the Outbox pattern, though more complex in Ruby).
- **Deployment:** Use tools like Docker, `systemd`, or process managers (like Foreman or Puma with background workers) to manage the consumer process lifecycle.
- **Monitoring:** Integrate monitoring tools (e.g., Prometheus exporters, Datadog agents) to track consumer lag, processing rates, and error counts.

## Conclusion

Ruby, paired with the `ruby-kafka` gem, offers a productive and expressive way to build services that participate in a Kafka-based event-driven architecture. Its dynamic nature and rich ecosystem make it particularly well-suited for tasks like data aggregation, analytics, and rapid prototyping, as demonstrated by our Analytics service.

In this chapter, we covered:

- Configuring the `ruby-kafka` client.
- Subscribing to topics and processing messages using `consumer.each_message`.
- Integrating Protobuf for cross-language event serialization.
- Running the consumer concurrently with other tasks.
- Basic error handling and offset management concepts in `ruby-kafka`.

This concludes our exploration of the specific language implementations in Part III. We've seen how Go, Kotlin, and Ruby each bring their unique strengths to the table, all effectively integrated through Kafka as the central event backbone. In Part IV, we will shift our focus to higher-level architectural patterns

commonly used in event-driven systems, starting with managing data contracts using schemas.

## Chapter 10: Contracts & Schemas with Protocol Buffers

In a polyglot event-driven system, one of the most critical challenges is ensuring that services written in different languages can reliably communicate with each other. As we transition from Part III, where we explored language-specific implementations, to Part IV, where we examine higher-level architectural patterns, we begin with the foundation of inter-service communication: data contracts and schemas.

This chapter explores how Protocol Buffers (Protobuf) provides a language-agnostic way to define, evolve, and enforce data contracts across our polyglot services. We'll examine the practical implementation in our reference system, covering schema design principles, versioning strategies, and integration with Kafka.

### The Need for Strong Contracts in Event-Driven Systems

In traditional request-response architectures, API contracts are often defined using specifications like OpenAPI (formerly Swagger), which document the endpoints, request/response formats, and validation rules. In event-driven architectures, we need a similar mechanism for defining the structure and semantics of events flowing through the system.

Without well-defined contracts, several problems can arise:

#### 1. Misinterpretation of Data

When a Go service produces an event that a Ruby service consumes, how does the Ruby service know what fields to expect and how to interpret them? Without a shared understanding, the consumer might misinterpret the data, leading to subtle bugs that are difficult to detect and debug.

#### 2. Brittle Coupling

If producers and consumers rely on implicit, undocumented contracts, changes to event structures can easily break downstream services. This creates a form of brittle coupling that undermines one of the key benefits of event-driven architecture: the ability to evolve services independently.

#### 3. Validation Challenges

Without a formal schema, validating incoming events becomes a manual, error-prone process. Each service must implement its own validation logic, leading to inconsistencies and potential security vulnerabilities.

## 4. Documentation Overhead

Maintaining separate documentation for event structures across multiple languages creates significant overhead and increases the risk of documentation drift, where the actual implementation diverges from the documented contract.

## Enter Protocol Buffers

Protocol Buffers (Protobuf), developed by Google, is a language-neutral, platform-neutral, extensible mechanism for serializing structured data. It addresses the challenges above by providing:

1. **Language-Agnostic Schema Definition:** Define data structures once, generate code for multiple languages.
2. **Efficient Serialization:** Compact binary format that's faster and smaller than alternatives like JSON or XML.
3. **Schema Evolution:** Built-in mechanisms for backward and forward compatibility.
4. **Strong Typing:** Compile-time type checking helps catch errors early.
5. **Documentation:** Self-documenting schemas with support for comments.

## Protobuf Basics

A Protobuf schema is defined in a `.proto` file using a simple, language-independent syntax. Here's a simplified example from our reference implementation:

```
// From kafka_content/idl/order.proto
syntax = "proto3";

package order;

option go_package = "github.com/scrappybara/kafka-watermill/idl/go/order";
option java_package = "com.scrappybara.kw.idl";
option java_outer_classname = "OrderProto";
option java_multiple_files = true;
option ruby_package = "KafkaWatermill.IDL";

// Event representing an order created by a user
message OrderCreated {
    string order_id = 1;
    string user_id = 2;
    repeated OrderItem items = 3;
    float total_amount = 4;
    string timestamp = 5;
}

// Supporting message type
message OrderItem {
    string product_id = 1;
    string name = 2;
}
```

```

    int32 quantity = 3;
    float price = 4;
}

```

This schema defines two message types: `OrderCreated` and `OrderItem`. The `OrderCreated` message includes a list (**repeated**) of `OrderItem` messages, demonstrating composition. Each field has a type (e.g., `string`, `int32`, `float`) and a unique field number (e.g., 1, 2, 3) that identifies the field in the binary encoding.

The `option` directives specify language-specific settings, such as package names and class names, ensuring that the generated code follows idiomatic conventions for each target language.

## Implementing Protobuf in Our Polyglot System

Let's examine how our reference implementation uses Protobuf to define and enforce contracts across Go, Kotlin, and Ruby services.

### 1. Centralized Schema Repository

All `.proto` files are stored in a central location (`kafka_content/idl/`), serving as the single source of truth for event definitions. This centralization ensures consistency and makes it easier to review and evolve schemas over time.

```

kafka_content/
  idl/
    order.proto      # Defines order-related events
    ...              # Other domain-specific schemas

```

### 2. Code Generation for Multiple Languages

The Protobuf compiler (`protoc`) generates language-specific code from the `.proto` files. This generated code includes classes/structs for each message type, serialization/deserialization methods, and utility functions.

For our polyglot system, we generate code for Go, Kotlin (Java), and Ruby:

```

# Generate Go code
protoc --proto_path=idl --go_out=go/gen order.proto

# Generate Java code (for Kotlin)
protoc --proto_path=idl --java_out=kotlin/src/main/java order.proto

# Generate Ruby code
protoc --proto_path=idl --ruby_out=ruby/lib order.proto

```

The generated code follows the package/namespace conventions specified in the `option` directives, ensuring idiomatic integration with each language.

### 3. Integration with Kafka Producers and Consumers

Each service uses the generated Protobuf code to serialize events before publishing them to Kafka and deserialize events when consuming from Kafka.

## Go (with Watermill)

```
// Publishing an event
orderCreated := &order.OrderCreated{
    OrderId:    uuid.New().String(),
    UserId:     userID,
    Items:      mapToProtoItems(items),
    TotalAmount: calculateTotal(items),
    Timestamp:  time.Now().Format(time.RFC3339),
}

// Serialize using Protobuf
payload, err := proto.Marshal(orderCreated)
if err != nil {
    return err
}

// Create Watermill message
msg := message.NewMessage(uuid.New().String(), payload)
msg.Metadata.Set("content-type", "application/protobuf")
msg.Metadata.Set("event-type", "OrderCreated")

// Publish to Kafka
return publisher.Publish("order.created", msg)

// Consuming an event
func HandleOrderCreated(msg *message.Message) ([]*message.Message, error) {
    // Deserialize using Protobuf
    var orderCreated order.OrderCreated
    if err := proto.Unmarshal(msg.Payload, &orderCreated); err != nil {
        return nil, fmt.Errorf("failed to unmarshal OrderCreated: %w", err)
    }

    // Process the event
    log.Printf("Received order: %s for user: %s with %d items",
        orderCreated.OrderId, orderCreated.UserId, len(orderCreated.Items))

    // ... business logic ...
}
```

## Kotlin (with Spring Kafka)

```
// Publishing an event
val orderCreated = OrderCreated.newBuilder()
    .setOrderId(UUID.randomUUID().toString())
    .setUserId(userID)
    .addAllItems(items.map { mapToProtoItem(it) })
    .setTotalAmount(calculateTotal(items))
    .setTimestamp(Instant.now().toString())
    .build()
```



```

// Serialize using Protobuf
val payload = orderCreated.toByteArray()

// Publish to Kafka
kafkaTemplate.send("order.created", orderCreated.orderId, payload)

// Consuming an event
@KafkaListener(topics = ["order.created"])
fun handleOrderCreated(payload: ByteArray) {
    // Deserialize using Protobuf
    val orderCreated = OrderCreated.parseFrom(payload)

    // Process the event
    logger.info("Received order: ${orderCreated.orderId} for user: ${orderCreated.userId}")

    // ... business logic ...
}

```

### Ruby (with ruby-kafka)

```

# Publishing an event (less common in our reference implementation)
order_created = KafkaWatermill::IDL::OrderCreated.new(
  order_id: SecureRandom.uuid,
  user_id: user_id,
  items: items.map { |item| map_to_proto_item(item) },
  total_amount: calculate_total(items),
  timestamp: Time.now.iso8601
)

# Serialize using Protobuf
payload = KafkaWatermill::IDL::OrderCreated.encode(order_created)

# Publish to Kafka
producer.produce(payload, topic: "order.created", key: order_created.order_id)

# Consuming an event
consumer.each_message do |message|
  case message.topic
  when "order.created"
    # Deserialize using Protobuf
    order_created = KafkaWatermill::IDL::OrderCreated.decode(message.value)

    # Process the event
    logger.info("Received order: #{order_created.order_id} for user: #{order_created.user_id}")

    # ... business logic ...
  end
end
end

```

## 4. Metadata and Content Type

In addition to the serialized payload, it's often useful to include metadata about the event, such as its type and serialization format. This metadata can be included in Kafka message headers or, as in our Watermill example, in message metadata.

```
// Go example with Watermill
msg := message.NewMessage(uuid.New().String(), payload)
msg.Metadata.Set("content-type", "application/protobuf")
msg.Metadata.Set("event-type", "OrderCreated")
```

This metadata helps consumers determine how to deserialize the message and can be used for routing, filtering, and monitoring.

## Schema Design Principles

Designing effective Protobuf schemas requires careful consideration of several factors. Here are key principles we follow in our reference implementation:

### 1. Domain-Driven Design

Our schemas reflect the domain model, with message types corresponding to domain events like `OrderCreated`, `PaymentProcessed`, and `OrderFulfilled`. This alignment with the domain model makes the schemas more intuitive and maintainable.

### 2. Event-First Design

Rather than starting with internal data models and exposing them as events, we design our events specifically for inter-service communication. This “event-first” approach ensures that events contain exactly the information needed by consumers, no more and no less.

### 3. Explicit Naming

We use explicit, descriptive names for message types and fields, avoiding abbreviations and ambiguous terms. This makes the schemas self-documenting and reduces the risk of misinterpretation.

```
// Good: Explicit naming
message OrderCreated {
    string order_id = 1;
    string user_id = 2;
    repeated OrderItem items = 3;
    float total_amount = 4;
    string timestamp = 5;
}
```

```
// Avoid: Ambiguous naming
message Order {
    string id = 1;
    string uid = 2;
```

```

    repeated Item i = 3;
    float total = 4;
    string ts = 5;
}

```

#### 4. Composition Over Inheritance

Protobuf doesn't support inheritance, but it does support composition through nested messages and imports. We use composition to create reusable components like `OrderItem` that can be included in multiple message types.

```

// Reusable component
message OrderItem {
    string product_id = 1;
    string name = 2;
    int32 quantity = 3;
    float price = 4;
}

// Used in multiple message types
message OrderCreated {
    // ... other fields ...
    repeated OrderItem items = 3;
}

message OrderFulfilled {
    // ... other fields ...
    repeated OrderItem items = 4;
}

```

#### 5. Minimal Dependencies

We minimize dependencies between schema files to reduce coupling and make evolution easier. When dependencies are necessary, we use imports to include only what's needed.

```

// Importing only what's needed
import "common/address.proto";

message OrderShipped {
    string order_id = 1;
    common.Address shipping_address = 2;
    // ... other fields ...
}

```

### Schema Evolution and Compatibility

One of the most powerful features of Protobuf is its support for schema evolution while maintaining compatibility. This is crucial in event-driven systems, where producers and consumers may be updated independently.

## Backward and Forward Compatibility

- **Backward Compatibility:** New schema can read data written with old schema.
- **Forward Compatibility:** Old schema can read data written with new schema.

Protobuf supports both types of compatibility through careful field numbering and default values.

## Compatibility Rules

To maintain compatibility when evolving schemas, follow these rules:

1. **Never change field numbers:** Field numbers are used in the binary encoding, so changing them would break compatibility.
2. **Never change field types:** Changing a field from `string` to `int32`, for example, would cause deserialization errors.
3. **Never remove required fields:** In proto3, all fields are optional by default, but removing a field that consumers expect can still cause issues.
4. **Add new fields with care:** New fields should have default values that make sense when old consumers encounter them.
5. **Use reserved fields and tags:** When removing fields, mark them as reserved to prevent future reuse.

```
message OrderCreated {  
    // Existing fields  
    string order_id = 1;  
    string user_id = 2;  
    repeated OrderItem items = 3;  
    float total_amount = 4;  
    string timestamp = 5;  
  
    // New field added in v2  
    string currency = 6;  
  
    // Reserved fields from removed features  
    reserved 7, 8, 9;  
    reserved "promotion_code", "referral_source";  
}
```

## Versioning Strategies

There are several approaches to versioning Protobuf schemas:

1. **Implicit Versioning** With implicit versioning, we rely on Protobuf's compatibility features without explicitly marking versions. New fields are added, and consumers are designed to handle missing fields gracefully.

This approach works well for minor changes but can become difficult to manage for significant schema changes.

**2. Explicit Versioning in Message Names** Another approach is to include version numbers in message names:

```
message OrderCreatedV1 {  
    // V1 fields  
}  
  
message OrderCreatedV2 {  
    // V2 fields, possibly including all V1 fields  
}
```

This approach makes versions explicit but requires producers to know which version to use and may lead to code duplication.

**3. Versioned Packages** A third approach is to version at the package level:

```
// v1/order.proto  
package order.v1;  
  
message OrderCreated {  
    // V1 fields  
}  
  
// v2/order.proto  
package order.v2;  
  
message OrderCreated {  
    // V2 fields  
}
```

This approach provides clear separation between versions but requires more complex import management.

**Our Approach: Evolutionary Design with Careful Documentation** In our reference implementation, we primarily use implicit versioning with careful documentation of changes. Each schema file includes a version history in comments, and we follow strict compatibility rules when making changes.

```
// order.proto  
// Version History:  
// v1.0.0 (2023-01-15): Initial version  
// v1.1.0 (2023-03-22): Added 'currency' field to OrderCreated  
// v1.2.0 (2023-06-10): Added 'shipping_method' field to OrderFulfilled  
  
syntax = "proto3";  
// ... rest of the schema ...
```

For major breaking changes, we would consider more explicit versioning strategies, but our focus on backward compatibility has made this unnecessary so far.

## Beyond Basic Protobuf: Advanced Techniques

While basic Protobuf usage covers most needs, our reference implementation employs several advanced techniques to enhance our schema management.

### 1. Custom Scalar Types

Protobuf's built-in scalar types (string, int32, etc.) are often sufficient, but sometimes we need more specific types. For example, we might want to ensure that a string field contains a valid UUID or timestamp.

We address this through validation in the application code and clear document (Content truncated due to size limit. Use line ranges to read in chunks)

## Chapter 11: Sagas for Distributed Transactions

In distributed systems, maintaining data consistency across multiple services is a significant challenge. Traditional ACID (Atomicity, Consistency, Isolation, Durability) transactions, common in monolithic applications with a single database, are often impractical or impossible to implement across independent microservices. As discussed in Chapter 2, event-driven architectures typically embrace eventual consistency. However, many business processes require a higher degree of coordination and atomicity than simple eventual consistency provides. This is where the Saga pattern comes into play.

This chapter delves into the Saga pattern, a fundamental technique for managing distributed transactions in event-driven systems. We explore how sagas use a sequence of local transactions coordinated through events to ensure that a complex business process either completes successfully across all involved services or is correctly compensated if any step fails.

### The Problem: Atomic Operations Across Services

Consider our e-commerce order processing workflow:

1. **Order Service:** Creates an order.
2. **Payment Service:** Processes the payment.
3. **Inventory Service:** Reserves the items.
4. **Shipping Service:** Arranges shipment.

Each of these steps involves a local transaction within the respective service (e.g., updating a database). However, the overall business process (placing and fulfilling an order) requires all these steps to succeed logically. What happens if payment succeeds, but inventory reservation fails because an item is out of stock?

In a monolithic system, a distributed transaction coordinator (like JTA) might be used, or all operations might occur within a single database transaction. In a microservices architecture, these options are generally unavailable or undesirable due to tight coupling and performance implications.

We need a way to ensure that if any step fails, the effects of the preceding successful steps are undone or compensated for, effectively rolling back the business

transaction.

## The Saga Pattern: A Sequence of Local Transactions

The Saga pattern addresses this challenge by breaking down a distributed transaction into a sequence of local transactions, each executed by a single service. Each local transaction updates the service's own data and publishes an event indicating its outcome.

If a step succeeds, the saga proceeds to the next step by triggering it with the published event. If a step fails, the saga executes a series of **compensating transactions** to undo the work done by the preceding successful steps.

### Key Characteristics of Sagas:

- **Sequence of Local Transactions:** Each step is atomic within its service boundary.
- **Event-Driven Coordination:** Events trigger subsequent steps or compensating actions.
- **Compensating Transactions:** For each step that can succeed, there must be a corresponding compensating transaction to undo its effects.
- **Eventual Consistency:** Sagas achieve atomicity at the business process level, but the system may be temporarily inconsistent while the saga is in progress or compensating.

## Saga Implementation Approaches: Choreography vs. Orchestration

There are two primary ways to coordinate the steps in a saga:

### 1. Choreography: Events Trigger Actions

In a choreographed saga, there is no central coordinator. Each service listens for events from other services and reacts accordingly. If a service completes its local transaction, it publishes an event. Other services listen for this event and trigger the next step in the saga.

#### Example (Order Processing - Choreography):

1. **Order Service:** Creates order -> Publishes **OrderCreated** event.
2. **Payment Service:** Listens for **OrderCreated** -> Processes payment -> Publishes **PaymentProcessed** event.
3. **Inventory Service:** Listens for **PaymentProcessed** -> Reserves inventory -> Publishes **InventoryReserved** event.
4. **Shipping Service:** Listens for **InventoryReserved** -> Arranges shipment -> Publishes **ShipmentArranged** event.

**Compensation (Choreography):** If Inventory Service fails to reserve items, it publishes an **InventoryReservationFailed** event. Payment Service listens for this and initiates a refund (compensation for **PaymentProcessed**). Order Service listens and marks the order as failed (compensation for **OrderCreated**).

graph LR

```
A[Order Service] -- OrderCreated --> B((Kafka))
```

```

B -- OrderCreated --> C[Payment Service]
C -- PaymentProcessed --> B
B -- PaymentProcessed --> D[Inventory Service]
D -- InventoryReserved --> B
B -- InventoryReserved --> E[Shipping Service]
E -- ShipmentArranged --> B

subgraph Compensation Flow
  D -- InventoryReservationFailed --> B
  B -- InventoryReservationFailed --> C
  C -- PaymentRefunded --> B
  B -- InventoryReservationFailed --> A
  A -- OrderFailed --> B
end

```

**Pros of Choreography:** - Simple: No central point of coordination. - Loose Coupling: Services only need to know about relevant events, not the entire workflow.

**Cons of Choreography:** - Difficult to Track: Understanding the overall state of the saga can be challenging as the logic is distributed across services. - Cyclic Dependencies: Can arise if services need to listen to events from services further down the chain. - Complexity Grows: Managing compensation logic across many services can become complex.

## 2. Orchestration: A Central Coordinator

In an orchestrated saga, a central coordinator (the orchestrator) manages the sequence of steps. The orchestrator sends commands to services to execute local transactions and listens for reply events indicating the outcome.

The orchestrator maintains the state of the saga and decides the next step based on the replies. If a step fails, the orchestrator explicitly commands the preceding services to execute their compensating transactions.

### Example (Order Processing - Orchestration):

1. **Orchestrator:** Receives `CreateOrder` command -> Sends `ProcessPayment` command to Payment Service.
2. **Payment Service:** Processes payment -> Publishes `PaymentProcessed` event.
3. **Orchestrator:** Listens for `PaymentProcessed` -> Sends `ReserveInventory` command to Inventory Service.
4. **Inventory Service:** Reserves inventory -> Publishes `InventoryReserved` event.
5. **Orchestrator:** Listens for `InventoryReserved` -> Sends `ArrangeShipment` command to Shipping Service.
6. **Shipping Service:** Arranges shipment -> Publishes `ShipmentArranged` event.
7. **Orchestrator:** Listens for `ShipmentArranged` -> Marks saga as complete.

**Compensation (Orchestration):** If Inventory Service publishes `InventoryReservationFailed`,



the Orchestrator receives it and sends a `RefundPayment` command to Payment Service and a `MarkOrderFailed` command to Order Service.

```
graph LR
    O[Saga Orchestrator] -- ProcessPaymentCmd --> C[Payment Service]
    C -- PaymentProcessedEvt --> O
    O -- ReserveInventoryCmd --> D[Inventory Service]
    D -- InventoryReservedEvt --> O
    O -- ArrangeShipmentCmd --> E[Shipping Service]
    E -- ShipmentArrangedEvt --> O

    subgraph Compensation Flow
        D -- InventoryReservationFailedEvt --> O
        O -- RefundPaymentCmd --> C
        O -- MarkOrderFailedCmd --> A[Order Service]
    end
end
```

**Pros of Orchestration:** - Centralized Logic: The entire workflow is defined in one place, making it easier to understand and manage. - Explicit State Management: The orchestrator explicitly tracks the saga's state. - Simpler Service Logic: Services only need to execute commands and publish reply events; they don't need to know about the overall saga flow.

**Cons of Orchestration:** - Central Point of Failure: The orchestrator can become a bottleneck or single point of failure (though it can be made resilient). - Tighter Coupling (potentially): Services are coupled to the orchestrator's commands.

## Implementing Saga Orchestration in Kotlin

Our reference implementation uses the **orchestration** approach for the shipping process, implemented within the Kotlin Shipping service (`kafka_content/kotlin-service`). Let's examine the key components.

### Saga Data (`Saga.kt`, `ShippingSaga.kt`)

We need a way to store the state associated with each saga instance. This includes the input data, intermediate results, current status, and any error information.

```
// From kafka_content/kotlin-service/src/main/kotlin/com/scrapybara/kw/shipping/saga/Saga.kt
enum class SagaStatus {
    STARTED, COMPLETED, FAILED, COMPENSATING, COMPENSATION_COMPLETED
}

interface SagaData {
    val sagaId: String
    val startTime: Instant
    var endTime: Instant?
    var currentStatus: SagaStatus
    var compensating: Boolean
    var error: String?
```

```

}

// From kafka_content/kotlin-service/src/main/kotlin/com/scrapybara/kw/shipping/saga/ShippingSaga.kt
data class ShippingSagaData(
    override val sagaId: String = UUID.randomUUID().toString(),
    override val startTime: Instant = Instant.now(),
    override var endTime: Instant? = null,
    override var currentStatus: SagaStatus = SagaStatus.STARTED,
    override var compensating: Boolean = false,
    override var error: String? = null,

    // Shipping-specific data
    val orderId: String,
    val allItemsAvailable: Boolean,
    var shipmentId: String? = null,
    var trackingNumber: String? = null,
    var notificationSent: Boolean = false
) : SagaData

```

The `ShippingSagaData` class holds all the information needed throughout the shipping saga, including the initial `orderId` and `allItemsAvailable` flag, as well as results from subsequent steps like `shipmentId` and `trackingNumber`.

### Saga Steps (Saga.kt)

Each step in the saga is defined with its forward action (handler) and its compensating action.

```

// From kafka_content/kotlin-service/src/main/kotlin/com/scrapybara/kw/shipping/saga/Saga.kt
class SagaStep<T> {
    val name: String,
    // Handler function: takes current data, performs action, returns updated data
    val handler: suspend (T) -> T,
    // Compensation function: takes current data, performs undo action, returns updated data
    val compensation: suspend (T) -> T
}

```

### Saga Coordinator (Saga.kt)

The `SagaCoordinator` class provides a generic mechanism to execute a list of `SagaSteps`, manage state, and handle compensation.

```

// From kafka_content/kotlin-service/src/main/kotlin/com/scrapybara/kw/shipping/saga/Saga.kt
class SagaCoordinator<T : SagaData> {
    private val sagaName: String,
    private val steps: List<SagaStep<T>>,
    private val kafkaTemplate: KafkaTemplate<String, ByteArray> // For publishing audit events
} {
    private val logger = LoggerFactory.getLogger(SagaCoordinator::class.java)

    suspend fun executeSaga(data: T): T {
        try {

```

```

publishSagaEvent(data, "STARTED")
var currentData = data

// Execute forward steps
steps.forEachIndexed { index, step ->
    try {
        currentData = step.handler(currentData)
        publishStepEvent(currentData, step.name, "COMPLETED")
    } catch (e: Exception) {
        // Error occurred, initiate compensation
        publishStepEvent(currentData, step.name, "FAILED", e.message)
        currentData.compensating = true
        currentData.error = e.message
        currentData.currentStatus = SagaStatus.COMPENSATING
        publishSagaEvent(currentData, "COMPENSATING")

        // Execute compensation steps in reverse
        for (i in index - 1 downTo 0) {
            val compensationStep = steps[i]
            try {
                currentData = compensationStep.compensation(currentData)
                publishStepEvent(currentData, compensationStep.name, "COMPENSATION_COMPLETED")
            } catch (ce: Exception) {
                // Log compensation error, but continue compensating
                publishStepEvent(currentData, compensationStep.name, "COMPENSATION_FAILED", ce.message)
            }
        }

        currentData.currentStatus = SagaStatus.COMPENSATION_COMPLETED
        publishSagaEvent(currentData, "COMPENSATION_COMPLETED")
        throw e // Re-throw original error
    }
}

// Saga completed successfully
currentData.currentStatus = SagaStatus.COMPLETED
publishSagaEvent(currentData, "COMPLETED")
return currentData

} catch (e: Exception) {
    // Saga failed (either during forward or compensation)
    data.currentStatus = SagaStatus.FAILED
    publishSagaEvent(data, "FAILED", e.message)
    throw e
}

}

// Helper methods to publish saga/step audit events to Kafka
private fun publishSagaEvent(data: T, status: String, error: String? = null) { /* ... */
private fun publishStepEvent(data: T, stepName: String, status: String, error: String? = null) { /* ... */

```

```
}
```

This coordinator iterates through the defined steps. If a `handler` fails, it iterates backward through the preceding steps, executing their `compensation` functions. It also publishes audit events (`SagaEvent`, `SagaStepEvent`) to Kafka topics (`saga.events`, `saga.step.events`) to provide visibility into the saga's progress.

### Defining the Shipping Saga (`ShippingSaga.kt`)

We define the specific steps for the shipping saga, injecting dependencies needed for each step (like `ShippingTrackerService`, `KafkaOperations`).

```
// From kafka_content/kotlin-service/src/main/kotlin/com/scrapybara/kw/shipping/saga/ShippingSaga.kt
@Component
class ShippingSagaManager(
    private val sagaCoordinatorFactory: SagaCoordinatorFactory,
    private val shippingTrackerService: ShippingTrackerService,
    private val kafkaOps: KafkaOperations // For publishing events like ShipmentCreated
) {
    private val logger = LoggerFactory.getLogger(ShippingSagaManager::class.java)

    private val shippingSagaCoordinator: SagaCoordinator<ShippingSagaData>

    init {
        val steps = listOf(
            SagaStep(
                name = "CreateShipment",
                handler = { data -> createShipmentHandler(data) },
                compensation = { data -> cancelShipmentCompensation(data) }
            ),
            SagaStep(
                name = "NotifyCustomer",
                handler = { data -> notifyCustomerHandler(data) },
                compensation = { data -> /* No compensation needed for notification */ data }
            )
            // Add more steps as needed (e.g., SchedulePickup, UpdateInventory)
        )
        shippingSagaCoordinator = sagaCoordinatorFactory.createCoordinator("ShippingSaga", steps)
    }

    suspend fun startShippingSaga(orderId: String, allItemsAvailable: Boolean): ShippingSagaData {
        val initialData = ShippingSagaData(
            orderId = orderId,
            allItemsAvailable = allItemsAvailable
        )

        // If items are not available, fail the saga immediately
        if (!allItemsAvailable) {
            logger.warn("Items not available for order $orderId, failing shipping saga early")
            initialData.currentStatus = SagaStatus.FAILED
            initialData.error = "Inventory not available"
        }
    }
}
```

```

        // publishSagaEvent(initialData, "FAILED", initialData.error) // Coordinator h
        // Potentially publish OrderCancelled event here or rely on another service
        return initialData // Or throw exception if coordinator expects it
    }

```

```

    return shipp

```

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## # Chapter 12: Event Sourcing

In the previous chapter, we explored how the Saga pattern helps maintain `data` consistency

Event Sourcing represents a paradigm shift `in` how applications store and manage `data`. Rath

In `this` chapter, we'll explore the principles of Event Sourcing, examine its implementatio

### ## Understanding Event Sourcing

#### ### The Traditional Approach: State-Based Persistence

In traditional applications, we typically store the current state of entities `in` a databas

#### orders

```

id: "order-123" customer_id: "cust-456" status: "shipped" total_amount:
99.99 created_at: "2023-06-15T10:30:00Z" updated_at: "2023-06-16T14:45:00Z"

```

When the order status changes (e.g., from "processing" to "shipped"), we update the `statu

1. **Loss of History**: We lose information about how the order reached its current state.
2. **Limited Auditability**: Without additional logging, it's difficult to track who made
3. **Temporal Queries**: Answering questions like "What was the status of this order yeste
4. **Concurrency Challenges**: Handling concurrent updates often relies on optimistic or p

#### ### The Event Sourcing Approach: Event-Based Persistence

Event Sourcing takes a fundamentally different approach. Instead of storing the current st

#### order\_\_events

```

id: "event-1", aggregate_id: "order-123", type: "OrderCreated", data: {...},
timestamp: "2023-06-15T10:30:00Z" id: "event-2", aggregate_id: "order-123",
type: "PaymentConfirmed", data: {...}, timestamp: "2023-06-15T10:35:00Z"
id: "event-3", aggregate_id: "order-123", type: "OrderProcessing", data: {...},
timestamp: "2023-06-15T11:00:00Z" id: "event-4", aggregate_id: "order-123",
type: "OrderShipped", data: {...}, timestamp: "2023-06-16T14:45:00Z"

```

The current state of the order is derived by replaying all events from the beginning. This

1. **Complete History**: We retain the full history of how the order evolved over time.
2. **Built-in Auditability**: Each event captures what happened, when it happened, and oft
3. **Temporal Queries**: We can reconstruct the state of the order at any point in time by
4. **Simplified Concurrency**: Since events are append-only, many concurrency issues are e

### Key Concepts in Event Sourcing

Before diving into implementation details, let's clarify some key concepts:

#### Events

Events are immutable records of something that happened in the past. They are typically na

```
```go
// Example event structure
type OrderShipped struct {
    BaseEvent          // Common fields like AggregateID, Timestamp
    TrackingNumber string
    ShippingMethod string
    EstimatedDelivery time.Time
}
```

**Aggregates** An aggregate is a cluster of domain objects that can be treated as a single unit. In Event Sourcing, aggregates are the entities whose state is managed through events. Each aggregate has a unique identifier, and all events related to that aggregate reference this identifier.

```
// Example aggregate structure
type Order struct {
    BaseAggregate          // Common fields and methods
    ID string
    CustomerID string
    Items []OrderItem
    Status string
    TotalAmount float64
    // ... other fields
}
```

**Event Store** The event store is the persistence mechanism for events. It provides methods to save new events and retrieve the event stream for a specific aggregate. In our Kafka-based system, Kafka itself can serve as an event store, with topics organized by aggregate type.

**Snapshots** As the number of events for an aggregate grows, replaying all events to reconstruct the current state can become inefficient. Snapshots provide a performance optimization by periodically capturing the aggregate's state, allowing us to start replay from the snapshot rather than from the beginning.

## Implementing Event Sourcing in Go

Our reference implementation includes a Go package for Event Sourcing (`kafka_content/pkg/eventsourcing`). Let's examine its key components.

### Event and Aggregate Interfaces

The package defines interfaces for events and aggregates:

```
// From kafka_content/pkg/eventsourcing/eventsourcing.go  
// Event is the base interface for all events  
type Event interface {  
    GetAggregateID() string  
    GetEventType() string  
    GetTimestamp() time.Time  
}  
  
// Aggregate is the base interface for event-sourced aggregates  
type Aggregate interface {  
    GetID() string  
    ApplyEvent(event Event) error  
    GetUncommittedEvents() []Event  
    ClearUncommittedEvents()  
}
```

These interfaces establish the contract for events and aggregates in our system. Events must provide methods to access their aggregate ID, type, and timestamp. Aggregates must be able to apply events, track uncommitted events, and provide access to their ID.

### Base Implementations

The package provides base implementations of these interfaces that can be embedded in concrete types:

```
// From kafka_content/pkg/eventsourcing/eventsourcing.go  
// BaseEvent provides common functionality for all events  
type BaseEvent struct {  
    AggregateID string `json:"aggregate_id"`  
    EventType   string `json:"event_type"`  
    Timestamp   time.Time `json:"timestamp"`  
}  
  
// BaseAggregate provides common functionality for all aggregates  
type BaseAggregate struct {  
    ID string  
    uncommittedEvents []Event  
    mutex sync.Mutex  
}
```

The `BaseEvent` struct implements the `Event` interface, providing common fields and methods. Similarly, `BaseAggregate` implements most of the `Aggregate` interface, managing uncommitted events and providing thread-safe access.

## Event Store Interface and Implementation

The package defines an interface for event stores and provides a simple in-memory implementation:

```
// From kafka_content/pkg/eventsourcing/eventsourcing.go
// EventStore is the interface for storing and retrieving events
type EventStore interface {
    SaveEvents(aggregateID string, events []Event) error
    GetEvents(aggregateID string) ([]Event, error)
}

// InMemoryEventStore is a simple in-memory implementation of EventStore
type InMemoryEventStore struct {
    events map[string] []Event
    mutex  sync.RWMutex
}
```

In a production system, we would implement this interface using Kafka or another persistent store. The in-memory implementation is useful for testing and demonstration purposes.

## Repository Pattern

The package uses the Repository pattern to provide a higher-level API for loading and saving aggregates:

```
// From kafka_content/pkg/eventsourcing/eventsourcing.go
// Repository is the interface for loading and saving aggregates
type Repository[T Aggregate] interface {
    Load(id string) (T, error)
    Save(aggregate T) error
}

// GenericRepository is a generic implementation of Repository
type GenericRepository[T Aggregate] struct {
    eventStore      EventStore
    aggregateFactory func(id string) T
    eventHandlerMap map[string]func(T, Event) error
}
```

The `GenericRepository` uses Go's generics to provide a type-safe way to work with different aggregate types. It takes an event store, a factory function for creating new aggregates, and a map of event handlers for applying different event types to the aggregate.

## Loading and Saving Aggregates

The repository's `Load` method reconstructs an aggregate by replaying its events:

```
// From kafka_content/pkg/eventsourcing/eventsourcing.go
// Load loads an aggregate from the event store
func (r *GenericRepository[T]) Load(id string) (T, error) {
```



```

// Create a new aggregate instance
aggregate := r.aggregateFactory(id)

// Get all events for this aggregate
events, err := r.eventStore.GetEvents(id)
if err != nil {
    return aggregate, fmt.Errorf("error loading events: %w", err)
}

// Apply each event to the aggregate
for _, event := range events {
    if err := aggregate.ApplyEvent(event); err != nil {
        return aggregate, fmt.Errorf("error applying event: %w", err)
    }
}

return aggregate, nil
}

```

The Save method persists uncommitted events to the event store:

```

// From kafka_content/pkg/eventsourcing/eventsourcing.go
// Save saves uncommitted events to the event store
func (r *GenericRepository[T]) Save(aggregate T) error {
    // Get uncommitted events
    events := aggregate.GetUncommittedEvents()
    if len(events) == 0 {
        return nil
    }

    // Save events to the event store
    if err := r.eventStore.SaveEvents(aggregate.GetID(), events); err != nil {
        return fmt.Errorf("error saving events: %w", err)
    }

    // Clear uncommitted events
    aggregate.ClearUncommittedEvents()

    return nil
}

```

## Using Event Sourcing in Our Order Service

Now that we understand the core components of our Event Sourcing implementation, let's see how they're used in the Order service.

### Defining Order Events

First, we define the events that can occur in the order lifecycle:

```

// Example event definitions (simplified)
type OrderCreated struct {

```

```

    eventsourcing.BaseEvent
    CustomerID    string
    Items         []OrderItem
    TotalAmount   float64
}

type PaymentProcessed struct {
    eventsourcing.BaseEvent
    PaymentID      string
    Amount         float64
    Status         string // "success" or "failed"
    TransactionID  string
}

type OrderShipped struct {
    eventsourcing.BaseEvent
    TrackingNumber  string
    ShippingMethod  string
    EstimatedDelivery time.Time
}

// ... other event types

```

Each event type embeds `BaseEvent` to inherit common functionality and adds fields specific to that event type.

## Implementing the Order Aggregate

Next, we implement the Order aggregate, which knows how to apply different event types:

```

// Example Order aggregate (simplified)
type Order struct {
    eventsourcing.BaseAggregate
    CustomerID    string
    Items         []OrderItem
    Status        string
    TotalAmount   float64
    PaymentID     string
    TrackingNumber string
    // ... other fields
}

// ApplyEvent applies an event to the order
func (o *Order) ApplyEvent(event eventsourcing.Event) error {
    switch e := event.(type) {
    case *OrderCreated:
        return o.applyOrderCreated(e)
    case *PaymentProcessed:
        return o.applyPaymentProcessed(e)
    case *OrderShipped:

```

```

        return o.applyOrderShipped(e)
    // ... other event types
    default:
        return fmt.Errorf("unknown event type: %s", event.GetEventType())
    }
}

// Event-specific apply methods
func (o *Order) applyOrderCreated(e *OrderCreated) error {
    o.ID = e.GetAggregateID()
    o.CustomerID = e.CustomerID
    o.Items = e.Items
    o.TotalAmount = e.TotalAmount
    o.Status = "created"
    return nil
}

func (o *Order) applyPaymentProcessed(e *PaymentProcessed) error {
    o.PaymentID = e.PaymentID
    if e.Status == "success" {
        o.Status = "payment_confirmed"
    } else {
        o.Status = "payment_failed"
    }
    return nil
}

func (o *Order) applyOrderShipped(e *OrderShipped) error {
    o.TrackingNumber = e.TrackingNumber
    o.Status = "shipped"
    return nil
}

```

The `ApplyEvent` method uses a type switch to delegate to event-specific apply methods. Each apply method updates the aggregate's state based on the event's data.

## Creating and Modifying Orders

To create a new order, we instantiate the aggregate, create an event, apply it to the aggregate, and save it:

```

// Example: Creating a new order
func CreateOrder(repo eventsourcing.Repository[*Order], customerID string, items []OrderItem) error {
    // Generate a new order ID
    orderID := uuid.New().String()

    // Create a new order aggregate
    order := &Order{
        BaseAggregate: eventsourcing.BaseAggregate{ID: orderID},
    }
}

```

```

// Create the OrderCreated event
event := &OrderCreated{
    BaseEvent: eventsourcing.BaseEvent{
        AggregateID: orderID,
        EventType:   "OrderCreated",
        Timestamp:   time.Now(),
    },
    CustomerID: customerID,
    Items:      items,
    TotalAmount: calculateTotal(items),
}

// Apply the event to the aggregate
if err := order.ApplyEvent(event); err != nil {
    return nil, err
}

// Add the event to uncommitted events
order.AddEvent(event)

// Save the aggregate (which saves the uncommitted events)
if err := repo.Save(order); err != nil {
    return nil, err
}

return order, nil
}

```

To modify an existing order, we load it, create and apply a new event, and save it:

```

// Example: Processing payment for an order
func ProcessPayment(repo eventsourcing.Repository[*Order], orderID string, paymentID string) (*Order, error) {
    // Load the order
    order, err := repo.Load(orderID)
    if err != nil {
        return nil, err
    }

    // Create the PaymentProcessed event
    event := &PaymentProcessed{
        BaseEvent: eventsourcing.BaseEvent{
            AggregateID: orderID,
            EventType:   "PaymentProcessed",
            Timestamp:   time.Now(),
        },
        PaymentID: paymentID,
        Amount:    amount,
        Status:    status,
    }
}

```

```

    // Apply the event to the aggregate
    if err := order.ApplyEvent(event); err != nil {
        return nil, err
    }

    // Add the event to uncommitted events
    order.AddEvent(event)

    // Save the aggregate (which saves the uncommitted events)
    if err := repo.Save(order); err != nil {
        return nil, err
    }

    return order, nil
}

```

## Integrating with Kafka

In a Kafka-based system, we can implement the `EventStore` interface using Kafka as the underlying storage mechanism:

```

// KafkaEventStore implements EventStore using Kafka
type KafkaEventStore struct {
    producer message.Publisher
    consumer message.Subscriber
    topicName string
}

// SaveEvents publishes events to Kafka
func (s *KafkaEventStore) SaveEvents(aggregateID string, events []eventsourcing.Event) error {
    for _, event := range events {
        // Serialize the event (e.g., using Protocol Buffers or JSON)
        payload, err := serializeEvent(event)
        if err != nil {
            return err
        }

        // Create a Watermill message
        msg := message.NewMessage(uuid.New().String(), payload)
        msg.Metadata.Set("aggregate_id", aggregateID)
        msg.Metadata.Set("event_type", event.GetEventType())

        // Publish to Kafka
        if err := s.producer.Publish(s.topicName, msg); err != nil {
            return err
        }
    }
}

```

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## # Chapter 13: CQRS (Command Query Responsibility Segregation)

In the previous chapter, we explored Event Sourcing, a pattern that captures all changes to

CQRS is an architectural pattern that separates the model used for updating data (commands

This chapter delves into the CQRS pattern, explaining its principles, benefits, and challenges.

### ## The Problem: One Model to Rule Them All?

Traditional application architectures often use a single data model for both commands (writing

1. **Executing Commands**: Loading an object, modifying its state based on business rules
2. **Executing Queries**: Retrieving objects based on various criteria, potentially joining

While simple for basic CRUD (Create, Read, Update, Delete) applications, this unified model

- **Conflicting Requirements**: The optimal model for enforcing business rules during writes
- **Performance Bottlenecks**: Complex queries on the write model can impact write performance
- **Scalability Issues**: Scaling the single model to handle both read and write loads efficiently
- **Complexity Creep**: The single model can become overly complex, trying to satisfy both

### ## CQRS: Separating Commands and Queries

CQRS addresses these issues by explicitly separating the responsibilities of handling commands

1. **Command Model**: Responsible for processing commands, enforcing business rules and idempotency
2. **Query Model (Read Model)**: Responsible for providing data needed for queries and data consistency

Crucially, these two models can use entirely different representations and even different

```
```mermaid
graph LR
    User --> C[Command API]
    User --> Q[Query API]

    subgraph Command Side
        C --> CH[Command Handler]
        CH --> Agg[Aggregate / Write Model]
        Agg --> DB_Write[(Write Database / Event Store)]
    end

    subgraph Query Side
        Q --> QH[Query Handler]
        QH --> DB_Read[(Read Database / Query Model)]
    end

    DB_Write -- Events / Data Sync --> Sync[Synchronization Logic]
```

Sync --> DB\_Read

## Key Components of a CQRS Architecture

- **Commands:** Imperative messages representing an intent to change the system state (e.g., `CreateOrderCommand`, `ProcessPaymentCommand`). Commands are typically handled by a single handler.
- **Command Handlers:** Process commands, interact with the command model (e.g., load aggregates, validate rules), and persist changes.
- **Command Model (Write Model):** The model used by command handlers, often involving aggregates and domain logic. It focuses on consistency and enforcing business rules.
- **Events:** Notifications that something significant has happened as a result of a command being processed (e.g., `OrderCreated`, `PaymentProcessed`). Events are the primary mechanism for synchronizing the query model.
- **Event Handlers (Projectors):** Listen for events published by the command model and update the query model accordingly. These handlers are responsible for transforming event data into the desired read model format.
- **Query Model (Read Model):** A data model specifically designed and optimized for querying. It can be denormalized and tailored to specific read use cases. There can be multiple query models for different purposes.
- **Queries:** Messages representing a request for data (e.g., `GetOrderByIdQuery`, `FindOrdersByCustomerQuery`). Queries do not change the system state.
- **Query Handlers:** Process queries by retrieving data directly from the query model.

## CQRS and Event Sourcing: A Powerful Combination

CQRS and Event Sourcing are distinct patterns, but they complement each other exceptionally well:

- **Event Sourcing as the Write Model:** When using Event Sourcing, the event store naturally serves as the persistence mechanism for the command model. Aggregates are loaded by replaying events, commands generate new events, and these events are appended to the store.
- **Events Drive Read Model Updates:** The stream of events produced by the event-sourced command model is the perfect input for updating the query model(s). Event handlers (projectors) can subscribe to the event stream and build or update the read models asynchronously.

```
graph LR
    User --> C[Command API]
    User --> Q[Query API]

    subgraph Command Side (Event Sourced)
        C --> CH[Command Handler]
        CH --> Agg[Aggregate]
        Agg -- Loads Events --> ES[(Event Store / Kafka)]
        CH -- Appends New Events --> ES
    end
end
```

```

subgraph Query Side
  Q --> QH[Query Handler]
  QH --> RM[(Read Model Database)]
end

ES -- Events --> EH[Event Handler / Projector]
EH --> RM

```

In this combined architecture:

1. Commands are processed by loading the relevant aggregate from its event history.
2. Business logic is executed, potentially generating new events.
3. New events are appended to the event store (e.g., a Kafka topic).
4. Event handlers (projectors) consume these events from the event store.
5. Projectors update one or more specialized read models (e.g., relational tables, document databases, search indexes) based on the event data.
6. Queries are served directly and efficiently from these optimized read models.

## Implementing CQRS in Our Polyglot System

While our reference implementation doesn't explicitly showcase separate read models updated by projectors, we can outline how CQRS could be applied, leveraging Kafka as the event bus.

### Command Side

- **Aggregates:** The `Order` aggregate (potentially event-sourced as described in Chapter 12) resides on the command side, handling commands like `CreateOrder`, `ProcessPayment`, `ShipOrder`.
- **Command Handlers:** Reside within services like the Order Service or Payment Service. They load aggregates, validate commands, execute business logic, and generate events.
- **Event Store:** Kafka topics (e.g., `order.events`) serve as the event store, storing the sequence of events generated by the command handlers.

### Query Side

We could introduce dedicated read models optimized for specific query needs:

1. **Order Summary Read Model:** A denormalized view stored, perhaps, in a document database (like MongoDB) or a relational table, containing key information needed for displaying order lists or summaries (e.g., order ID, customer name, status, total amount, creation date).
2. **Product Catalog Read Model:** A read model optimized for searching and displaying product information, potentially stored in a search engine like Elasticsearch.
3. **Analytics Read Model:** The Ruby Analytics service already acts as a form of read model, aggregating data from various events for reporting purposes. This could be formalized further.



## Synchronization via Kafka

Kafka acts as the backbone connecting the command and query sides:

1. **Event Publication:** When the command side processes a command and generates events (e.g., `OrderCreated`, `PaymentProcessed`), these events are published to relevant Kafka topics.
2. **Event Consumption (Projection):** Dedicated event handlers (projectors), potentially running as separate services or within existing services, subscribe to these Kafka topics.
3. **Read Model Updates:** Each projector processes the events it receives and updates its corresponding read model. For example:
  - An `OrderSummaryProjector` listens to `OrderCreated`, `PaymentProcessed`, `OrderShipped`, etc., and updates the `OrderSummary` documents/rows.
  - An `AnalyticsProjector` (like our Ruby service) listens to various events and updates aggregated metrics.

### Example: Order Summary Projection

Let's imagine an `OrderSummaryProjector` implemented in Kotlin using Spring Kafka:

```
// Simplified Projector Example
@Service
class OrderSummaryProjector(
    private val orderSummaryRepository: OrderSummaryRepository // Repository for the read model
) {
    private val logger = LoggerFactory.getLogger(OrderSummaryProjector::class.java)

    @KafkaListener(topics = ["order.events"], groupId = "order-summary-projector-group")
    fun handleOrderEvent(payload: ByteArray, @Header(KafkaHeaders.RECEIVED_MESSAGE_KEY) key: String) {
        try {
            // Deserialize based on event type (could use headers or a wrapper envelope)
            // For simplicity, assume we can determine type
            when (val event = deserializeEvent(payload)) {
                is OrderCreated -> {
                    logger.info("Projecting OrderCreated: ${event.orderId}")
                    val summary = OrderSummary(
                        orderId = event.orderId,
                        customerId = event.customerId,
                        status = "CREATED",
                        totalAmount = event.totalAmount,
                        createdAt = event.timestamp
                        // ... other fields
                    )
                    orderSummaryRepository.save(summary)
                }
                is PaymentProcessed -> {
                    logger.info("Projecting PaymentProcessed: ${event.orderId}")
                    orderSummaryRepository.updateStatus(event.orderId,

```

```

        if (event.status == "success") "PAYMENT_CONFIRMED" else "PAYMENT_F
    }
    is OrderShipped -> {
        logger.info("Projecting OrderShipped: ${event.orderId}")
        orderSummaryRepository.updateStatusAndTracking(event.orderId, "SHIPPED
    }
    // ... handle other relevant events
}
} catch (e: Exception) {
    logger.error("Error projecting event for key $key", e)
    // Implement error handling (e.g., DLQ)
}
}

// Placeholder for deserialization logic
private fun deserializeEvent(payload: ByteArray): Any {
    // Implement logic to determine event type and deserialize (e.g., using Protobuf)
    // This might involve checking headers or using a schema registry
    return Any() // Replace with actual deserialized event
}

// Simplified Read Model Repository Interface
interface OrderSummaryRepository {
    fun save(summary: OrderSummary)
    fun updateStatus(orderId: String, status: String)
    fun updateStatusAndTracking(orderId: String, status: String, trackingNumber: String)
    // ... query methods like findById, findByCustomerId, etc.
}

// Simplified Read Model Data Class
data class OrderSummary(
    val orderId: String,
    val customerId: String,
    var status: String,
    val totalAmount: Float,
    val createdAt: String,
    var trackingNumber: String? = null
    // ... other denormalized fields
)

```

This projector listens to the `order.events` topic. When an event arrives, it deserializes it, determines the type, and updates the `OrderSummary` read model accordingly using a dedicated repository. Queries for order summaries would then interact directly with this `OrderSummaryRepository`.

## Benefits of CQRS

- **Optimized Models:** Allows tailoring the command model for consistency and the query model(s) for specific read performance needs.

- **Scalability:** Read and write workloads can be scaled independently. You can add more instances to handle read traffic without impacting write operations, and vice-versa.
- **Flexibility:** Easier to introduce new ways of querying data by adding new read models without affecting the command side.
- **Technology Choice:** Allows using different persistence technologies best suited for each model (e.g., Event Store/Kafka for writes, document DB/search index/relational DB for reads).
- **Separation of Concerns:** Clear separation between command processing logic and query logic.

## Challenges of CQRS

- **Increased Complexity:** Managing separate models, synchronization logic, and potentially different databases adds complexity compared to a single-model approach.
- **Eventual Consistency:** The query model is typically updated asynchronously based on events, meaning there's a delay between a write occurring and the change being reflected in reads. Applications must be designed to handle this potential staleness.
- **Infrastructure Overhead:** Requires infrastructure for event handling and potentially managing multiple databases or data stores.
- **Code Duplication (Potentially):** Some data structures might be duplicated (though potentially in different forms) between the command and query models.
- **Synchronization Logic:** Building reliable and resilient event handlers (projectors) to update read models requires careful design, including handling errors, retries, and idempotency.

## Conclusion

CQRS is a powerful pattern for managing the inherent differences between command and query operations in complex systems. By separating the models responsible for writes and reads, CQRS enables optimized performance, scalability, and flexibility.

When combined with Event Sourcing, CQRS provides a robust architecture where the event stream from the command side naturally drives the updates to one or more specialized read models. Kafka serves as an excellent backbone for this architecture, facilitating the asynchronous flow of events from the event store to the projectors that build the query models.

While CQRS introduces complexity and the challenge of eventual consistency, the benefits often outweigh the drawbacks in systems with demanding read/write requirements or complex domain logic. Understanding CQRS is crucial for designing scalable and maintainable event-driven microservices.

In the next chapter, we will shift our focus to a critical aspect of building reliable systems: Testing Event-Driven Architectures.

## Chapter 14: Testing Event-Driven Systems

Building robust and reliable event-driven systems requires a comprehensive testing strategy. The asynchronous, distributed nature of these architectures introduces unique challenges compared to testing traditional monolithic applications. Simply testing individual services in isolation is often insufficient; we need to verify the interactions between services, the correctness of event processing, and the end-to-end behavior of business workflows.

This chapter explores various testing techniques applicable to event-driven systems, particularly those built around Kafka. We will discuss unit testing, integration testing (including approaches using embedded Kafka and test containers), contract testing, and end-to-end testing, drawing examples from our reference implementation and best practices highlighted in books like “Kafka in Action” and “Kafka Streams in Action.”

### The Challenges of Testing Event-Driven Architectures

Testing event-driven systems presents several unique hurdles:

1. **Asynchronicity:** Interactions happen asynchronously via events. Verifying outcomes requires waiting for events to be processed, which can make tests slower and more complex.
2. **Distributed State:** Business processes often span multiple services, making it hard to track the overall state and verify consistency.
3. **Infrastructure Dependencies:** Services rely on the message broker (Kafka) and potentially other infrastructure components (databases, external APIs), which need to be managed during testing.
4. **Non-Determinism:** Factors like network latency, broker availability, and consumer group rebalancing can introduce non-determinism, making tests potentially flaky.
5. **Complex Failure Modes:** Failures can occur in various places – producers, consumers, the broker itself, or downstream services – requiring tests to cover these scenarios.

### Levels of Testing

A robust testing strategy employs multiple levels of testing, each focusing on different aspects of the system:

#### 1. Unit Testing

Unit tests focus on verifying the smallest testable parts of an application, typically individual functions or classes, in isolation from their dependencies.

**Scope:** Test specific business logic within a service, such as: - Command handlers validating commands. - Aggregate methods applying events and enforcing invariants. - Event handler (projector) logic transforming events into read models. - Utility functions and data transformations.

**Techniques:** Dependencies like repositories, Kafka producers/consumers, or external services are typically mocked or stubbed.

### Example (Go - Testing Aggregate Logic):

```
// order_test.go
func TestOrder_ApplyOrderCreated(t *testing.T) {
    order := &Order{BaseAggregate: eventsourcing.BaseAggregate{ID: "order-1"}}
    event := &OrderCreated{
        BaseEvent: eventsourcing.BaseEvent{AggregateID: "order-1"},
        CustomerID: "cust-1",
        Items:      []OrderItem{{ProductID: "prod-a", Quantity: 1, Price: 10.0}},
        TotalAmount: 10.0,
    }

    err := order.ApplyEvent(event)

    assert.NoError(t, err)
    assert.Equal(t, "order-1", order.ID)
    assert.Equal(t, "cust-1", order.CustomerID)
    assert.Equal(t, "created", order.Status)
    assert.Equal(t, 10.0, order.TotalAmount)
    assert.Len(t, order.Items, 1)
}
```

**Benefits:** Fast, isolated, easy to write and maintain. **Limitations:** Doesn't verify interactions between components or with infrastructure.

## 2. Integration Testing

Integration tests verify the interaction between different components within a service or between a service and its direct infrastructure dependencies (like Kafka or a database).

**Scope:** Test how components work together, such as: - A command handler interacting with an aggregate and an event store. - A Kafka consumer processing a message and updating a database. - A Kafka producer successfully sending a message to the broker.

**Techniques:** Often involve using test doubles for external services but real instances (or test-specific versions) of infrastructure like Kafka and databases.

**Testing Kafka Integration** Testing Kafka interactions requires a running Kafka broker. Common approaches include:

- **Embedded Kafka:** Libraries like `spring-kafka-test` (Java/Kotlin) or custom setups allow running an in-memory Kafka broker within the test process.
- **Testcontainers:** A popular library that provides lightweight, disposable instances of common infrastructure (including Kafka) running in Docker containers managed by the test framework.
- **Shared Test Cluster:** Using a dedicated Kafka cluster for testing (less common due to potential interference between tests).

**Example (Kotlin - Using Embedded Kafka with `spring-kafka-test`):**

Our reference implementation includes `KafkaIntegrationTest.kt`, which uses `@EmbeddedKafka` from `spring-kafka-test`:

```
// From kafka_content/kotlin-service/src/test/kotlin/com/scrapybara/kw/shipping/KafkaInteg
@ExtendWith(SpringExtension::class)
@SpringBootTest
@EmbeddedKafka(partitions = 1, topics = ["inventory.checked"]) // Starts embedded Kafka
class KafkaIntegrationTest {

    @Autowired
    private lateinit var kafkaTemplate: KafkaTemplate<String, Any> // For sending test mes

    @MockBean // Mock the actual service logic to isolate Kafka consumption
    private lateinit var shippingSagaManager: ShippingSagaManager

    @Test
    fun `should consume inventory checked event`() {
        // Given: An event to send
        val event = OrderProto.InventoryChecked.newBuilder()
            .setOrderId("test-order-123")
            .setAllItemsAvailable(true)
            // ... set other fields
            .build()

        // When: Send the event to the embedded Kafka topic
        kafkaTemplate.send("inventory.checked", "test-order-123", event.toByteArray())
            .get(10, TimeUnit.SECONDS) // Wait for send to complete

        // Then: Verify the consumer (ShippingService) called the mocked dependency
        verify(shippingSagaManager, timeout(5000)) // Use timeout for async verification
            .startShippingSaga(eq("test-order-123"), eq(true))
    }
}
```

This test: 1. Uses `@EmbeddedKafka` to start a Kafka broker before tests run. 2. Injects a `KafkaTemplate` configured to talk to the embedded broker. 3. Mocks the `ShippingSagaManager` to verify that the Kafka listener in `ShippingService` correctly invokes it after consuming the message. 4. Sends a test event to the `inventory.checked` topic. 5. Uses Mockito's `verify` with a `timeout` to assert that the mocked `startShippingSaga` method was called asynchronously by the listener.

#### Example (Using Testcontainers):

```
// Java example using Testcontainers
@Testcontainers
class KafkaServiceIntegrationTest {

    @Container
    static KafkaContainer kafka = new KafkaContainer(DockerImageName.parse("confluentinc/c

    // ... setup Kafka producers/consumers pointing to kafka.getBootstrapServers() ...
}
```

```

@Test
void shouldProduceAndConsumeMessage() {
    // Setup producer and consumer connected to the containerized Kafka
    // ...

    // Produce a message
    // ...

    // Consume the message and assert
    // ...
}
}

```

Testcontainers provide a more realistic environment than embedded Kafka, as they use the actual Kafka Docker image.

**Benefits:** Verify interactions with real infrastructure, catch integration issues early. **Limitations:** Slower than unit tests, require managing test infrastructure, may not cover interactions between multiple services.

### 3. Contract Testing

Contract tests focus on verifying the compatibility of interactions between services (or between a service and its consumers/providers) without requiring full end-to-end deployment.

**Scope:** Ensure that: - A service consuming events from Kafka can correctly deserialize and process events produced by another service according to a shared contract (e.g., Protobuf schema). - A service producing events adheres to the agreed-upon schema expected by its consumers.

**Techniques:** Tools like Pact or Spring Cloud Contract can be used. - **Consumer-Driven Contracts:** The consumer defines its expectations (the contract) for the events it needs. The provider (producer) then verifies that it meets this contract. - **Schema Registry:** Using a schema registry (like Confluent Schema Registry) enforces schema compatibility at runtime, acting as a form of contract enforcement.

#### Example (Conceptual - Using Schema Registry):

1. **Define Schema:** Define the `OrderCreated` event schema using Protobuf (as in Chapter 10).
2. **Register Schema:** Register the schema in the Confluent Schema Registry.
3. **Producer Test:** Write a test for the Order Service producer that:
  - Configures the Kafka producer to use the Schema Registry serializer.
  - Attempts to produce an `OrderCreated` event.
  - Verifies that the event is successfully serialized and registered against the schema.
4. **Consumer Test:** Write a test for a downstream service consumer (e.g., Payment Service) that:

- Configures the Kafka consumer to use the Schema Registry deserializer.
- Consumes a test message produced with the registered schema.
- Verifies that the event is correctly deserialized according to the expected schema.

**Benefits:** Ensures compatibility between services without full integration tests, allows services to evolve independently as long as contracts are met, faster feedback than end-to-end tests. **Limitations:** Doesn't verify the actual runtime behavior or business logic flow, requires maintaining contracts.

#### 4. End-to-End (E2E) Testing

E2E tests verify complete business workflows that span multiple services.

**Scope:** Simulate user interactions or system triggers and verify the final outcome across the entire distributed system. - Example: Test the complete order fulfillment process from placing an order via an API gateway to verifying the order status is updated to “shipped” and a notification is sent.

**Techniques:** Requires deploying all involved services and their dependencies (Kafka, databases) in a test environment. Tests interact with the system's entry points (e.g., API gateways, message producers) and verify results by querying read models, checking database states, or observing side effects (like published events or notifications).

**Example (Conceptual):**

```
# Pseudocode for an E2E test
def test_order_fulfillment_happy_path():
    # 1. Setup: Ensure services (Order, Payment, Inventory, Shipping, Kafka) are running
    order_api = OrderServiceClient("http://api-gateway/orders")
    shipping_db = ShippingReadModelClient("shipping-db-test")
    notification_spy = NotificationSpyClient("notification-service-test")

    # 2. Action: Place a new order via the API
    order_data = {"customer_id": "cust-e2e", "items": [{"product_id": "prod-e2e", "quantity": 1}]}
    response = order_api.create_order(order_data)
    order_id = response["order_id"]

    # 3. Verification: Wait and check the expected outcomes
    # Wait for events to propagate and be processed (requires robust waiting strategy)
    time.sleep(30) # Simple sleep, better to poll or wait for specific events

    # Check shipping status in read model
    shipment_status = shipping_db.get_status(order_id)
    assert shipment_status == "SHIPPED"

    # Check if notification was sent (assuming a test spy)
    notifications = notification_spy.get_notifications(order_id)
    assert len(notifications) > 0
    assert notifications[0]["type"] == "shipment_created"
```



**Challenges of E2E Tests:** - **Complexity:** Setting up and maintaining the test environment is complex. - **Slow Execution:** Tests involve network communication and asynchronous processing, making them slow. - **Flakiness:** Prone to intermittent failures due to timing issues, network glitches, or environment instability. - **Debugging:** Pinpointing the root cause of a failure can be difficult as it could be in any service or interaction.

**Benefits:** Verify the system works as a whole from the user's perspective, provide high confidence in business workflows. **Limitations:** Slow, expensive, brittle, provide late feedback in the development cycle.

## Best Practices for Testing Kafka Applications

- **Use a Mix of Test Levels:** Rely heavily on unit and integration tests for fast feedback. Use contract tests to ensure compatibility. Employ a smaller number of E2E tests for critical business workflows.
- **Isolate Tests:** Ensure tests don't interfere with each other, especially when using shared resources. Use unique topic names, consumer group IDs, or containerized infrastructure per test suite.
- **Manage Test Data:** Have strategies for generating realistic test data and cleaning up state between tests.
- **Handle Asynchronicity:** Use appropriate mechanisms (e.g., polling, await libraries like Awaitility, Kafka consumer APIs) to wait for asynchronous operations to complete before making assertions.
- **Test Failure Scenarios:** Explicitly test how the system handles errors, retries, and compensation logic (e.g., in Sagas).
- **Test Kafka Configuration:** Verify that producer and consumer configurations (acks, retries, serializers, etc.) are correct and lead to the desired behavior.
- **Leverage Testing Libraries:** Utilize libraries like `spring-kafka-test`, `Testcontainers`, `kafka-streams-test-utils`, and mocking frameworks to simplify test setup and execution.

## Conclusion

Testing event-driven systems built with Kafka requires a multi-layered approach. Unit tests provide fast feedback on isolated logic, while integration tests verify interactions with Kafka and other infrastructure using tools like embedded Kafka or Testcontainers. Contract testing ensures compatibility between services, often leveraging schema registries. Finally, end-to-end tests validate complete business workflows across the distributed system, albeit with higher complexity and cost.

By understanding the challenges and applying the appropriate techniques at each level, we can build confidence in the correctness and reliability of our polyglot, event-driven applications. The key is to balance the scope, speed, and complexity of different test types to achieve comprehensive coverage efficiently.

In the next chapter, we will explore another crucial aspect of operating event-driven systems: Monitoring and Observability.

## Chapter 15: Monitoring & Observability

In event-driven systems, particularly those built around Kafka, understanding what’s happening inside your application is both crucial and challenging. The distributed, asynchronous nature of these systems means that traditional monitoring approaches often fall short. When a business process spans multiple services connected through events, pinpointing the source of issues, understanding performance bottlenecks, or simply confirming that everything is working correctly requires a comprehensive observability strategy.

This chapter explores the principles and practices of monitoring and observability for Kafka-based, event-driven systems. We’ll cover key metrics to track, logging strategies, distributed tracing, alerting, and dashboarding, with practical examples from our reference implementation and insights from industry best practices.

### Monitoring vs. Observability

Before diving into specifics, let’s clarify the distinction between monitoring and observability:

- **Monitoring** is about tracking and alerting on known, predefined metrics and states. It answers questions like “Is the system up?” or “Is the consumer lag exceeding our threshold?”
- **Observability** is a broader concept that enables understanding the internal state of a system through its external outputs. It helps answer questions you didn’t anticipate, like “Why is this particular order stuck in processing?” or “What’s the exact path this event took through our system?”

A mature system needs both: monitoring to alert you when things go wrong, and observability to help you understand why they went wrong and how to fix them.

### Key Metrics for Kafka-Based Systems

Effective monitoring starts with identifying the right metrics to track. For Kafka-based systems, these fall into several categories:

#### 1. Kafka Broker Metrics

These metrics help monitor the health and performance of the Kafka cluster itself:

- **Broker Uptime:** Time since the broker started.
- **Request Rate:** Number of requests per second (produce, fetch, meta-data).
- **Request Latency:** Time taken to process requests.
- **Disk Usage:** Available disk space and utilization.
- **Network I/O:** Bytes in/out per second.
- **Under-replicated Partitions:** Partitions with fewer than the configured number of replicas.

- **Offline Partitions:** Partitions without an active leader.
- **Active Controller Count:** Should be exactly 1 across the cluster.

## 2. Producer Metrics

These metrics help monitor the health and performance of your Kafka producers:

- **Record Send Rate:** Number of records sent per second.
- **Byte Rate:** Bytes sent per second.
- **Request Latency:** Time taken for the broker to acknowledge records.
- **Record Error Rate:** Number of records that failed to be sent.
- **Batch Size:** Average and max size of batches sent.
- **Record Queue Time:** Time records spend in the producer buffer.

## 3. Consumer Metrics

These metrics help monitor the health and performance of your Kafka consumers:

- **Record Consumption Rate:** Number of records consumed per second.
- **Byte Rate:** Bytes consumed per second.
- **Consumer Lag:** Number of messages the consumer is behind the producer (critical metric).
- **Poll Rate:** Number of poll calls per second.
- **Poll Latency:** Time taken to complete poll calls.
- **Commit Success/Failure Rate:** Rate of successful/failed offset commits.
- **Rebalance Rate:** Number of consumer group rebalances.

## 4. Application-Specific Metrics

Beyond Kafka-specific metrics, you should track metrics relevant to your business domain:

- **Event Processing Rate:** Number of events processed per second, by event type.
- **Event Processing Latency:** Time taken to process each event.
- **Business Process Completion Rate:** Rate at which end-to-end business processes (e.g., order fulfillment) complete.
- **Business Process Duration:** Time taken for end-to-end business processes.
- **Error Rates:** Number of errors encountered during event processing.
- **Dead Letter Queue (DLQ) Rate:** Rate at which messages are sent to DLQs.

### Example: Metrics in Our Kotlin Service

Our reference implementation includes metrics in the Kotlin Shipping service using Spring Boot Actuator and Micrometer:

```
// From kafka_content/kotlin-service/src/main/kotlin/com/scrapybara/kw/shipping/config/MetricsConfig.kt
@Configuration
class MetricsConfig(private val registry: MeterRegistry) {
```

```

@Bean
fun shippingMetrics(): ShippingMetrics {
    return ShippingMetrics(registry)
}

}

// Metrics collection class
class ShippingMetrics(private val registry: MeterRegistry) {

    // Counter metrics
    val kafkaMessagesReceived = registry.counter("shipping.kafka.messages.received")
    val kafkaMessagesFailed = registry.counter("shipping.kafka.messages.failed")
    val ordersProcessed = registry.counter("shipping.orders.processed")

    // Timer metric for measuring processing duration
    val shippingProcessingTime = registry.timer("shipping.processing.time")

    // Gauge metrics could be added for things like "current orders in processing"

    // Methods to increment counters with business context
    fun incrementOrdersFulfilled() {
        registry.counter("shipping.orders.fulfilled").increment()
    }

    fun incrementOrdersCancelled() {
        registry.counter("shipping.orders.cancelled").increment()
    }
}

```

These metrics are then used in the service code:

```

// From kafka_content/kotlin-service/src/main/kotlin/com/scrappybara/kw/shipping/ShippingService.kt
@Service
class ShippingService(
    // ... other dependencies
    private val metrics: ShippingMetrics,
    private val registry: MeterRegistry
) {

    @KafkaListener(topics = ["inventory.checked"], containerFactory = "kafkaListenerContainerFactory")
    fun handleInventoryChecked(payloadBytes: ByteArray) {
        metrics.kafkaMessagesReceived.increment()

        val sample = Timer.start(registry)
        try {
            val event = OrderProto.InventoryChecked.parseFrom(payloadBytes)

            // ... business logic ...

            if (event.allItemsAvailable) {

```

```

        metrics.incrementOrdersFulfilled()
    } else {
        metrics.incrementOrdersCancelled()
    }
    metrics.ordersProcessed.increment()

} catch (e: Exception) {
    metrics.kafkaMessagesFailed.increment()
    throw e
} finally {
    sample?.stop(metrics.shippingProcessingTime)
}
}
}

```

This approach captures: - The rate of Kafka messages received and failed - The rate of orders processed, fulfilled, and cancelled - The time taken to process shipping requests

These metrics can be exposed via Spring Boot Actuator's `/actuator/metrics` endpoint and collected by monitoring systems like Prometheus.

## Logging Strategies

While metrics provide aggregated numerical data, logs provide detailed contextual information about specific events and operations. Effective logging is crucial for debugging and understanding system behavior.

### Structured Logging

Traditional text-based logs are difficult to parse and analyze at scale. Structured logging formats like JSON make logs machine-readable while preserving human readability:

```

{
  "timestamp": "2023-06-15T14:30:00.123Z",
  "level": "INFO",
  "service": "shipping-service",
  "traceId": "abc123",
  "message": "Processing inventory checked event",
  "orderId": "order-456",
  "allItemsAvailable": true,
  "processingTimeMs": 45
}

```

### Contextual Information

Include relevant business context in logs to make them useful for debugging:

- **Correlation IDs:** Include trace and span IDs (discussed in the Distributed Tracing section).

- **Business Identifiers:** Include order IDs, customer IDs, or other business-relevant identifiers.
- **Event Metadata:** Include event types, Kafka topics, partitions, and offsets.
- **Performance Data:** Include processing times and resource usage where relevant.

## Log Levels

Use appropriate log levels to control verbosity:

- **ERROR:** Use for exceptions and errors that require immediate attention.
- **WARN:** Use for unusual but non-critical situations.
- **INFO:** Use for normal operational events (e.g., “Processing order XYZ”).
- **DEBUG:** Use for detailed information useful during development and debugging.
- **TRACE:** Use for very detailed information, typically only enabled during specific debugging sessions.

## Example: Logging in Our Go Service

Our reference implementation includes structured logging in the Go Order service using the `watermill` logger:

```
// Example from Go service (conceptual)
logger := watermill.NewStdLogger(false, false)

// Logging with context
logger.Info("Processing order created event", watermill.LogFields{
    "order_id":      event.OrderID,
    "customer_id":   event.CustomerID,
    "total_amount":  event.TotalAmount,
    "items_count":   len(event.Items),
    "kafka_topic":   msg.Metadata.Get("kafka_topic"),
    "kafka_partition": msg.Metadata.Get("kafka_partition"),
    "kafka_offset":  msg.Metadata.Get("kafka_offset"),
})

// Error logging
if err != nil {
    logger.Error("Failed to process order", err, watermill.LogFields{
        "order_id": event.OrderID,
        // Additional context
    })
}
```

## Distributed Tracing

In a distributed system where a single business transaction spans multiple services connected by Kafka events, understanding the end-to-end flow is challenging. Distributed tracing addresses this by tracking the journey of a request or

event through the entire system.

### Key Concepts

- **Trace:** Represents the end-to-end journey of a request or business transaction.
- **Span:** Represents a single operation within a trace (e.g., processing an event, making a database query).
- **Trace Context:** Metadata (trace ID, span ID, etc.) that's propagated between services to link spans together.

### Propagating Trace Context via Kafka

To maintain trace continuity across Kafka-based service boundaries:

1. **Producer Side:** Include trace context in Kafka message headers or metadata.
2. **Consumer Side:** Extract trace context from the message and continue the trace.

```
// Example: Propagating trace context in Kotlin with Spring Cloud Sleuth (conceptual)
@Service
class OrderService(private val kafkaTemplate: KafkaTemplate<String, ByteArray>) {

    fun createOrder(order: Order) {
        // Business logic...

        // Create OrderCreated event
        val event = OrderCreated(/* ... */)
        val payload = event.toByteArray()

        // Create Kafka message with trace headers
        val headers = KafkaHeaders()
        // Spring Cloud Sleuth automatically adds trace headers

        // Send to Kafka with headers
        kafkaTemplate.send("order.created", order.id, payload)
    }
}

@Service
class PaymentService {

    @KafkaListener(topics = ["order.created"])
    fun handleOrderCreated(payload: ByteArray,
        @Header(KafkaHeaders.RECEIVED_MESSAGE_KEY) key: String,
        @Header("X-B3-TraceId") traceId: String?,
        @Header("X-B3-SpanId") spanId: String?) {
        // Spring Cloud Sleuth automatically continues the trace using headers

        // Process the event...
    }
}
```

```
}  
}
```

## Visualizing Traces

Distributed tracing systems like Jaeger, Zipkin, or AWS X-Ray provide UIs to visualize traces:

- **Trace Timeline:** Shows the sequence and duration of spans.
- **Service Dependency Graph:** Shows how services interact.
- **Span Details:** Shows detailed metadata about each operation.

These visualizations help identify bottlenecks, errors, and the flow of requests through the system.

## Alerting and Dashboarding

Collecting metrics, logs, and traces is only valuable if you can act on this information. This requires effective alerting and dashboarding.

### Alerting Best Practices

- **Alert on Symptoms, Not Causes:** Alert on customer-impacting issues (e.g., high error rates, excessive latency) rather than internal implementation details.
- **Reduce Noise:** Avoid alert fatigue by only alerting on actionable issues.
- **Define Clear Thresholds:** Set appropriate thresholds based on historical data and business requirements.
- **Include Context:** Provide enough information in alerts to understand the issue without additional investigation.
- **Implement Escalation Policies:** Define who should be notified and when.

### Key Alerts for Kafka Systems

- **Consumer Lag:** Alert when consumers fall too far behind producers.
- **Error Rates:** Alert on elevated error rates in producers, consumers, or application code.
- **Dead Letter Queue:** Alert when messages are sent to DLQs.
- **Processing Latency:** Alert when event processing takes longer than expected.
- **Broker Health:** Alert on broker failures, under-replicated partitions, or disk space issues.

### Dashboarding

Effective dashboards provide at-a-glance visibility into system health and performance. Consider organizing dashboards by:

- **Service:** Dashboards for each service showing its specific metrics.
- **Business Process:** Dashboards tracking end-to-end business processes (e.g., order fulfillment).



- **Infrastructure:** Dashboards for Kafka brokers, databases, and other infrastructure.
- **User Experience:** Dashboards focused on customer-facing metrics.

### Example Dashboard Components

- **Kafka Health:** Broker status, under-replicated partitions, consumer lag.
- **Throughput:** Messages per second, bytes per second.
- **Latency:** Processing times, end-to-end latency.
- **Error Rates:** Failed messages, application errors.
- **Business Metrics:** Orders processed, payments completed, etc.

## Implementing Observability in Our Polyglot System

Let's explore how observability might be implemented across our polyglot system:

### Go Service with Prometheus and OpenTelemetry

```
// Conceptual example for Go service
import (
    "github.com/prometheus/client_golang/prometheus"
    "go.opentelemetry.io/otel"
    "go.opentelemetry.io/otel/trace"
)

// Prometheus metrics
var (
    ordersProcessed = prometheus.NewCounter(prometheus.CounterOpts{
        Name: "orders_processed_total",
        Help: "Total number of processed orders",
    })
    processingDuration = prometheus.NewHistogram(prometheus.HistogramOpts{
        Name: "order_processing_duration_seconds",
        Help: "Time taken to process orders",
        Buckets: prometheus.DefBuckets,
    })
)

func init() {
    prometheus.MustRegister(ordersProcessed, processingDuration)
}

// Handler with tracing and metrics
func HandleOrderCreated(msg *message.Message) ([]*message.Message, error) {
    // Extract trace context from message metadata
    ctx := extractTraceContext(msg)

    // Start a new span
    ctx, span := otel.Tracer("order-service").Start(ctx, "HandleOrderCreated")
```

```

defer span.End()

// Record start time for metrics
startTime := time.Now()

// Deserialize message
var orderCreated OrderCreated
if err := proto.Unmarshal(msg.Payload, &orderCreated); err != nil {
    span.RecordError(err)
    span.SetStatus(codes.Error, "Failed to unmarshal")
    return nil, err
}

// Add event details to span
span.SetAttributes(
    attribute.String("order.id", orderCreated.OrderId),
    attribute.Float64("order.amount", orderCreated.TotalAmount),
)

// Process the event...

// Update metrics
ordersProcessed.Inc()
processingDuration.Observe(time.Since(startTime).Seconds())

return []*message.Message{outMsg}, nil
}

```

## Kotlin Service with Micrometer and Spring Cloud Sleuth

```

// Conceptual example for Kotlin service
@Service
class PaymentService(
    private val
    (Content truncated due to size limit. Use line ranges to read in chunks)

```

### # Chapter 16: Scaling & Performance Tuning

As event-driven systems grow in complexity and usage, ensuring they perform efficiently at

Drawing on insights from "Kafka in Action" and "Kafka Streams in Action," we'll examine ho

### ## Understanding Kafka's Scalability Model

Before diving into specific optimizations, it's important to understand Kafka's fundamenta

#### ### Partitions as the Unit of Parallelism

As discussed in Chapter 5, partitions are the basic unit of parallelism in Kafka. Each par

1. **Producer Parallelism**: Multiple producers can write to different partitions of the same topic.
2. **Consumer Parallelism**: Multiple consumers in a consumer group can process different partitions of the same topic.

Topic A  
Partition 0

Consumer 1

Producer 1      Partition 1      Producer 2  
2      Partition 2      Consumer 2  
         Producer 3      Partition 3      Consumer 3

### ### Broker Distribution

Kafka distributes partitions across multiple brokers, allowing the cluster to scale horizontally.

## ## Scaling Kafka Clusters

### ### Determining the Right Cluster Size

The optimal Kafka cluster size depends on several factors:

- **Expected Throughput**: Messages per second and average message size.
- **Retention Requirements**: How long data needs to be retained.
- **Replication Factor**: Higher replication factors require more storage and network bandwidth.
- **Topic and Partition Count**: More partitions require more memory and can impact broker performance.

From "Kafka in Action," a general guideline is to start with 3-5 brokers for development/testing.

### ### Partition Planning

Choosing the right number of partitions is crucial for scalability:

- **Too Few Partitions**: Limits consumer parallelism and can create bottlenecks.
- **Too Many Partitions**: Increases overhead on brokers and can lead to resource contention.

Guidelines for determining partition count:

1. **Consider Consumer Scalability**: The maximum number of consumers in a group equals the number of partitions.
2. **Consider Throughput Requirements**: Each partition has a throughput limit based on the broker's capacity.
3. **Consider Message Key Distribution**: Ensure keys are distributed evenly across partitions for efficient lookups.
4. **Consider Broker Resources**: Each partition requires memory and file handles on the broker.

From "Kafka Streams in Action," a common starting point is to have at least as many partitions as consumers.

### ### Example Partition Calculation

Expected peak throughput: 100,000 messages/second  
Average message size: 1KB  
Expected consumer instances: 5 (initially), potentially growing to 10

get throughput per partition: 10,000 messages/second

Initial partition count =  $\text{Max}(\text{Expected consumers} \times 2, \text{Peak throughput} \div \text{Target throughput per partition}) = \text{Max}(5 \times 2, 100,000 \div 10,000) = \text{Max}(10, 10) = 10$  partitions

This calculation gives us 10 partitions, which accommodates our initial 5 consumers with r

## ## Optimizing Kafka Producers

Producer performance can significantly impact the overall throughput and latency of your s

### ### Batching and Compression

Batching multiple messages into a single request reduces network overhead and improves thr

```
```go
// Go example with Watermill (conceptual)
publisherConfig := kafka.PublisherConfig{
    Brokers:    []string{"kafka:9092"},
    Marshaler:  kafka.DefaultMarshaler{},
    // Enable batching
    PublishConfig: kafka.PublishConfig{
        Async:      true,
        BatchSize:  16384, // 16KB batch size
        BatchTimeout: time.Millisecond * 100,
        Compression: sarama.CompressionSnappy,
    },
}

// Kotlin example with Spring Kafka (conceptual)
@Bean
fun producerFactory(): ProducerFactory<String, ByteArray> {
    val configProps = mapOf(
        ProducerConfig.BOOTSTRAP_SERVERS_CONFIG to bootstrapServers,
        ProducerConfig.KEY_SERIALIZER_CLASS_CONFIG to StringSerializer::class.java,
        ProducerConfig.VALUE_SERIALIZER_CLASS_CONFIG to ByteArraySerializer::class.java,
        // Enable batching
        ProducerConfig.BATCH_SIZE_CONFIG to 16384,
        ProducerConfig.LINGER_MS_CONFIG to 100,
        // Enable compression
        ProducerConfig.COMPRESSION_TYPE_CONFIG to "snappy"
    )
    return DefaultKafkaProducerFactory(configProps)
}
```

## Acknowledgment Settings

The `acks` configuration controls the durability guarantees for produced messages:

- **acks=0**: No acknowledgment (highest throughput, no durability guarantee).
- **acks=1**: Leader acknowledgment (balanced throughput and durability).
- **acks=all**: All replicas acknowledgment (highest durability, lower throughput).

Choose based on your reliability requirements:

```
// Kotlin example with Spring Kafka
ProducerConfig.ACKS_CONFIG to "all" // For critical data
// or
ProducerConfig.ACKS_CONFIG to "1"    // For less critical, high-throughput data
```

### Asynchronous vs. Synchronous Sending

Asynchronous sending improves throughput by not waiting for broker acknowledgments before sending the next batch:

```
// Go example with Watermill
publisherConfig.PublishConfig.Async = true

// Kotlin example with Spring Kafka
kafkaTemplate.send("topic", key, payload)
    .addCallback(
        { result -> logger.debug("Message sent successfully") },
        { ex -> logger.error("Failed to send message", ex) }
    )
// Continue processing without waiting
```

### Buffer Memory and Retries

Configure buffer memory and retry settings based on your expected load and reliability requirements:

```
// Kotlin example with Spring Kafka
ProducerConfig.BUFFER_MEMORY_CONFIG to 33554432, // 32MB
ProducerConfig.RETRIES_CONFIG to 3,
ProducerConfig.RETRY_BACKOFF_MS_CONFIG to 100
```

## Optimizing Kafka Consumers

Consumer optimization is crucial for processing messages efficiently:

### Consumer Group Design

Design consumer groups to match your processing requirements:

- **One Consumer Group per Logical Service**: Each service that needs to process all messages should have its own consumer group.
- **Balance Partitions and Consumers**: Ideally, each consumer should handle a similar number of partitions.

```
Service A (Order Processing)
    Consumer Group A
```

```

        Consumer A1 (handles partitions 0, 1)
        Consumer A2 (handles partitions 2, 3)

Service B (Analytics)
    Consumer Group B
        Consumer B1 (handles partition 0)
        Consumer B2 (handles partition 1)
        Consumer B3 (handles partition 2)
        Consumer B4 (handles partition 3)

```

## Fetch Size and Max Poll Records

Configure how many records consumers fetch in each poll:

```

// Kotlin example with Spring Kafka
ConsumerConfig.MAX_POLL_RECORDS_CONFIG to 500,
ConsumerConfig.FETCH_MIN_BYTES_CONFIG to 1024,
ConsumerConfig.FETCH_MAX_BYTES_CONFIG to 52428800 // 50MB

```

## Consumer Concurrency

Adjust the number of concurrent consumers based on your processing needs:

```

// Kotlin example with Spring Kafka
@Bean
fun kafkaListenerContainerFactory(): ConcurrentKafkaListenerContainerFactory<String, ByteArray>() {
    val factory = ConcurrentKafkaListenerContainerFactory<String, ByteArray>()
    factory.consumerFactory = consumerFactory()
    factory.setConcurrency(3) // 3 consumer threads
    return factory
}

```

## Offset Commit Strategy

Choose an appropriate offset commit strategy:

- **Auto-commit:** Simplest but may lead to duplicate processing after crashes.
- **Manual commit:** More control but requires careful implementation.
- **Exactly-once processing:** Most complex but eliminates duplicates.

```

// Kotlin example with Spring Kafka - Manual commit
ConsumerConfig.ENABLE_AUTO_COMMIT_CONFIG to false,

// In the listener
@KafkaListener(topics = ["topic"], containerFactory = "kafkaListenerContainerFactory")
fun listen(record: ConsumerRecord<String, ByteArray>, acknowledgment: Acknowledgment) {
    try {
        // Process the record
        processRecord(record)

        // Commit the offset
    }
}

```

```

        acknowledgment.acknowledge()
    } catch (e: Exception) {
        // Handle error
        logger.error("Error processing record", e)
        // Decide whether to commit or not based on error type
    }
}

```

## Language-Specific Optimizations

Each language in our polyglot system has unique considerations for Kafka performance:

### Go (with Watermill)

Go's lightweight goroutines make it well-suited for concurrent processing:

- **Leverage Goroutines:** Use goroutines for parallel processing within a consumer.
- **Channel Buffering:** Configure appropriate channel buffer sizes to prevent blocking.
- **Memory Management:** Be mindful of memory usage in high-throughput scenarios.

```

// Example: Parallel processing in Go
router.AddHandler(
    "parallel_handler",
    "input.topic",
    subscriber,
    "output.topic",
    publisher,
    func(msg *message.Message) ([]*message.Message, error) {
        // Process in parallel using a worker pool
        results := make(chan *message.Message)
        errors := make(chan error)

        // Submit to worker pool
        workerPool.Submit(func() {
            result, err := processMessage(msg)
            if err != nil {
                errors <- err
                return
            }
            results <- result
        })

        // Wait for result or error
        select {
        case result := <-results:
            return []*message.Message{result}, nil
        case err := <-errors:

```

```

        return nil, err
    }
},
)

```

## Kotlin (with Spring Kafka)

Kotlin with Spring Kafka offers several optimization opportunities:

- **Coroutines:** Use coroutines for non-blocking I/O operations.
- **Batch Listeners:** Process multiple records in a single listener invocation.
- **Concurrent Listeners:** Configure multiple listener threads.

```

// Example: Batch listener with coroutines
@KafkaListener(topics = ["input.topic"], containerFactory = "batchListenerFactory")
fun listenBatch(records: List<ConsumerRecord<String, ByteArray>>, acknowledgment: Acknowledgeable) {
    runBlocking {
        val deferreds = records.map { record ->
            async(Dispatchers.IO) {
                processRecord(record)
            }
        }

        // Wait for all processing to complete
        deferreds.awaitAll()

        // Commit offsets for the entire batch
        acknowledgment.acknowledge()
    }
}

```

## Ruby (with ruby-kafka)

Ruby's GIL (Global Interpreter Lock) can limit concurrency, but there are still optimization options:

- **Multiple Processes:** Use multiple Ruby processes instead of threads.
- **Batch Processing:** Process multiple messages in each consumer loop iteration.
- **JRuby:** Consider JRuby to overcome GIL limitations for CPU-bound tasks.

```

# Example: Batch processing in Ruby
consumer.each_batch(max_bytes: 10_485_760, max_wait_time: 1.0) do |batch|
    # Process all messages in the batch
    batch.messages.each do |message|
        process_message(message)
    end

    # Commit offsets after processing the entire batch
    consumer.mark_message_as_processed(batch.messages.last)
end

```



```

        consumer.commit_offsets
    end

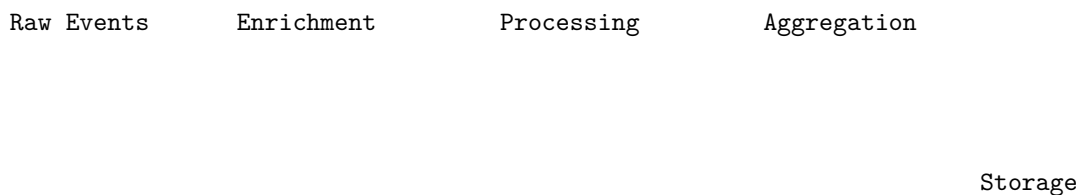
```

## Designing for High Throughput

Beyond individual optimizations, consider these architectural patterns for high-throughput systems:

### Parallel Processing Pipelines

Break complex processing into stages connected by Kafka topics:



This approach allows each stage to scale independently based on its specific processing requirements.

### Partitioning Strategies

Design partitioning strategies to maximize parallelism while maintaining ordering where needed:

- **Business Key Partitioning:** Ensure related events go to the same partition (e.g., partition by order ID).
- **Round-Robin Partitioning:** Maximize throughput when order doesn't matter.
- **Custom Partitioning:** Implement custom logic for complex scenarios.

```

// Kotlin example with Spring Kafka - Custom partitioner
@Bean
fun producerFactory(): ProducerFactory<String, ByteArray> {
    val configProps = mapOf(
        // ... other configs
        ProducerConfig.PARTITIONER_CLASS_CONFIG to CustomPartitioner::class.java
    )
    return DefaultKafkaProducerFactory(configProps)
}

class CustomPartitioner : Partitioner {
    override fun partition(
        topic: String,
        key: Any,
        keyBytes: ByteArray,
        value: Any,
        valueBytes: ByteArray,
    ) {

```

```

        cluster: Cluster
    ): Int {
        // Custom partitioning logic
        // ...
    }
}

```

## Caching and Local State

Maintain local caches or state stores to reduce external lookups:

```

// Kotlin example with Spring Kafka - Local cache
@Service
class OrderProcessor {
    private val orderCache = ConcurrentHashMap<String, Order>()

    @KafkaListener(topics = ["order.events"])
    fun processOrderEvent(record: ConsumerRecord<String, ByteArray>) {
        val orderId = record.key()

        // Check cache first
        val order = orderCache.computeIfAbsent(orderId) {
            // Only fetch from database if not in cache
            orderRepository.findById(it).orElse(null)
        }

        // Process event with cached order
        // ...
    }
}

```

## Handling Common Performance Challenges

### Slow Consumers

Slow consumers can lead to increasing lag and eventually consumer group rebalancing. Strategies to address this:

1. **Increase Consumer Parallelism:** Add more consumer in (Content truncated due to size limit. Use line ranges to read in chunks)

## Chapter 17: Security & Compliance

In event-driven architectures, particularly those built around Kafka, security and compliance considerations are paramount. As data flows through multiple services and is persisted in event streams, protecting that data from unauthorized access, ensuring its integrity, and meeting regulatory requirements become critical challenges. This chapter explores the security and compliance aspects of Kafka-based, polyglot event-driven systems, drawing on best practices from industry experts and our reference implementation.

## The Security Landscape for Event-Driven Systems

Event-driven architectures introduce unique security considerations compared to traditional request-response systems:

1. **Data Persistence:** Events are typically stored for extended periods, increasing the exposure window for sensitive data.
2. **Data Distribution:** The same data may be replicated across multiple topics, services, and data stores.
3. **Service Boundaries:** Events cross service boundaries, requiring careful access control.
4. **Polyglot Challenges:** Different languages and frameworks may have varying security capabilities and vulnerabilities.

## Authentication and Authorization in Kafka

### Authentication: Verifying Identity

Kafka supports several authentication mechanisms:

1. **SSL/TLS Client Authentication** Clients present certificates that are validated by the broker:

```
# Broker configuration
ssl.keystore.location=/var/private/ssl/kafka.server.keystore.jks
ssl.keystore.password=keystore-password
ssl.key.password=key-password
ssl.truststore.location=/var/private/ssl/kafka.server.truststore.jks
ssl.truststore.password=truststore-password
ssl.client.auth=required

# Client configuration
ssl.keystore.location=/var/private/ssl/client.keystore.jks
ssl.keystore.password=client-keystore-password
ssl.key.password=client-key-password
ssl.truststore.location=/var/private/ssl/client.truststore.jks
ssl.truststore.password=client-truststore-password
```

2. **SASL (Simple Authentication and Security Layer)** Kafka supports several SASL mechanisms:

- **PLAIN:** Simple username/password authentication (should only be used with TLS).
- **SCRAM:** Salted Challenge Response Authentication Mechanism, more secure than PLAIN.
- **GSSAPI:** Kerberos-based authentication.
- **OAUTHBEARER:** OAuth 2.0 token-based authentication.

```
# Broker configuration
listeners=SASL_SSL://kafka:9093
security.inter.broker.protocol=SASL_SSL
sasl.mechanism.inter.broker.protocol=PLAIN
```

```
sasl.enabled.mechanisms=PLAIN,SCRAM-SHA-256
```

```
# Client configuration
security.protocol=SASL_SSL
sasl.mechanism=SCRAM-SHA-256
sasl.jaas.config=org.apache.kafka.common.security.scram.ScramLoginModule required \
  username="client-user" \
  password="client-password";
```

### Authorization: Controlling Access

Once authenticated, authorization determines what operations clients can perform:

**1. ACL (Access Control Lists)** Kafka's built-in ACL system allows fine-grained control over topic operations:

```
# Grant read access to a consumer group
bin/kafka-acls.sh --bootstrap-server kafka:9093 \
  --command-config admin.properties \
  --add \
  --allow-principal User:payment-service \
  --consumer \
  --group payment-group \
  --topic order.events

# Grant write access to a producer
bin/kafka-acls.sh --bootstrap-server kafka:9093 \
  --command-config admin.properties \
  --add \
  --allow-principal User:order-service \
  --producer \
  --topic order.events
```

**2. Role-Based Access Control (RBAC)** Enterprise Kafka distributions like Confluent Platform offer RBAC for more sophisticated access control:

Role: OrderProcessor

Permissions:

- READ: Topic[order.events]
- WRITE: Topic[payment.events]
- CREATE: Topic[order.\*]

Assignments:

- Principal: User:order-service
- Principal: ServiceAccount:order-processor

### Implementing Authentication in Our Polyglot System

Each language in our polyglot system requires specific configuration for authentication:

### Go (with Watermill)

```
// Go example with Watermill and Sarama
saramaConfig := sarama.NewConfig()
saramaConfig.Net.SASL.Enable = true
saramaConfig.Net.SASL.Mechanism = sarama.SASLTypeSCRAMSHA256
saramaConfig.Net.SASL.User = "go-service"
saramaConfig.Net.SASL.Password = "go-service-password"
saramaConfig.Net.TLS.Enable = true
saramaConfig.Net.TLS.Config = &tls.Config{
    // TLS configuration
}

publisherConfig := kafka.PublisherConfig{
    Brokers:    []string{"kafka:9093"},
    Marshaler:  kafka.DefaultMarshaler{},
    OverwriteSaramaConfig: saramaConfig,
}

publisher, err := kafka.NewPublisher(publisherConfig, logger)
```

### Kotlin (with Spring Kafka)

```
// Kotlin example with Spring Kafka
@Bean
fun producerFactory(): ProducerFactory<String, ByteArray> {
    val configProps = mapOf(
        ProducerConfig.BOOTSTRAP_SERVERS_CONFIG to "kafka:9093",
        ProducerConfig.KEY_SERIALIZER_CLASS_CONFIG to StringSerializer::class.java,
        ProducerConfig.VALUE_SERIALIZER_CLASS_CONFIG to ByteArraySerializer::class.java,

        // Security configuration
        CommonClientConfigs.SECURITY_PROTOCOL_CONFIG to "SASL_SSL",
        SaslConfigs.SASL_MECHANISM to "SCRAM-SHA-256",
        SaslConfigs.SASL_JAAS_CONFIG to """
            org.apache.kafka.common.security.scram.ScramLoginModule required
            username="kotlin-service"
            password="kotlin-service-password";
        """.trimIndent(),

        // SSL configuration
        SslConfigs.SSL_TRUSTSTORE_LOCATION_CONFIG to "/path/to/truststore.jks",
        SslConfigs.SSL_TRUSTSTORE_PASSWORD_CONFIG to "truststore-password"
    )
    return DefaultKafkaProducerFactory(configProps)
}
```

### Ruby (with ruby-kafka)

```
# Ruby example with ruby-kafka
kafka = Kafka.new(
```

```

seed_brokers: ["kafka:9093"],
client_id: "ruby-service",

# SSL configuration
ssl_ca_cert: File.read("/path/to/ca.pem"),
ssl_client_cert: File.read("/path/to/client.pem"),
ssl_client_cert_key: File.read("/path/to/client.key"),

# SASL configuration
sasl_scram_username: "ruby-service",
sasl_scram_password: "ruby-service-password",
sasl_scram_mechanism: "sha256"
)

```

## Encryption and Data Protection

### Transport Encryption with TLS

TLS (Transport Layer Security) encrypts data in transit between clients and brokers:

```

# Broker configuration
listeners=SSL://kafka:9093
ssl.keystore.location=/path/to/kafka.server.keystore.jks
ssl.keystore.password=keystore-password
ssl.key.password=key-password
ssl.truststore.location=/path/to/kafka.server.truststore.jks
ssl.truststore.password=truststore-password

```

### Encryption at Rest

Kafka itself doesn't provide built-in encryption at rest, but several approaches can be used:

1. **Filesystem Encryption:** Encrypt the volumes where Kafka stores data.
2. **Application-Level Encryption:** Encrypt sensitive fields before producing messages.

```

// Kotlin example of application-level encryption
@Service
class EncryptionService(private val encryptionKey: SecretKey) {

    fun encrypt(plaintext: String): String {
        val cipher = Cipher.getInstance("AES/GCM/NoPadding")
        cipher.init(Cipher.ENCRYPT_MODE, encryptionKey)

        val iv = cipher.iv
        val encrypted = cipher.doFinal(plaintext.toByteArray())

        // Combine IV and encrypted data
        val result = ByteArray(iv.size + encrypted.size)
        System.arraycopy(iv, 0, result, 0, iv.size)
    }
}

```

```

        System.arraycopy(encrypted, 0, result, iv.size, encrypted.size)

        return Base64.getEncoder().encodeToString(result)
    }

    fun decrypt(ciphertext: String): String {
        val data = Base64.getDecoder().decode(ciphertext)

        // Extract IV and encrypted data
        val iv = data.copyOfRange(0, 12) // GCM IV is 12 bytes
        val encrypted = data.copyOfRange(12, data.size)

        val cipher = Cipher.getInstance("AES/GCM/NoPadding")
        val spec = GCMParameterSpec(128, iv)
        cipher.init(Cipher.DECRYPT_MODE, encryptionKey, spec)

        val decrypted = cipher.doFinal(encrypted)
        return String(decrypted)
    }
}

// Usage in a service
@Service
class PaymentService(private val encryptionService: EncryptionService) {

    fun processPayment(payment: Payment): PaymentProcessed {
        // Encrypt sensitive data
        val encryptedCardNumber = encryptionService.encrypt(payment.cardNumber)

        // Create event with encrypted data
        val event = PaymentProcessed(
            paymentId = payment.id,
            status = "success",
            encryptedCardData = encryptedCardNumber,
            // Other non-sensitive fields
            amount = payment.amount
        )

        return event
    }
}

```

## Data Masking and Tokenization

For sensitive data that must be stored in events:

1. **Data Masking:** Replace sensitive data with masked versions (e.g., "XXXX-XXXX-XXXX-1234").
2. **Tokenization:** Replace sensitive data with tokens that can be resolved to the original data only by authorized services.

```
// Kotlin example of tokenization
@Service
class TokenizationService(private val tokenRepository: TokenRepository) {

    fun tokenize(sensitiveData: String, dataType: String): String {
        // Generate a random token
        val token = UUID.randomUUID().toString()

        // Store the mapping
        tokenRepository.save(Token(
            token = token,
            sensitiveData = sensitiveData,
            dataType = dataType,
            createdAt = Instant.now()
        ))

        return token
    }

    fun detokenize(token: String): String? {
        return tokenRepository.findByToken(token)?.sensitiveData
    }
}
```

## Compliance Considerations

### Data Privacy Regulations

Event-driven systems must comply with regulations like GDPR, CCPA, and HIPAA:

**1. Right to Erasure (Right to be Forgotten)** Kafka's immutable log makes deleting specific records challenging. Strategies include:

- **Compacted Topics:** Use Kafka's log compaction to replace sensitive records with tombstones.
- **Encryption Key Rotation:** Encrypt sensitive data and delete the encryption keys when erasure is requested.
- **Data Anonymization:** Replace personal data with anonymized versions.

```
// Kotlin example of handling erasure requests
@Service
class GdprService(
    private val kafkaTemplate: KafkaTemplate<String, ByteArray>,
    private val encryptionKeyRepository: EncryptionKeyRepository
) {

    fun processErasureRequest(userId: String) {
        // 1. Publish a user deletion event
        val deletionEvent = UserDeletionRequested(
            userId = userId,

```



```

        requestedAt = Instant.now().toString()
    )
    kafkaTemplate.send("user.deletion.requested", userId, deletionEvent.toByteArray())

    // 2. Delete encryption keys for this user
    encryptionKeyRepository.deleteByUserId(userId)

    // 3. Log the erasure request for compliance
    logger.info("Erasure request processed for user: $userId")
}
}

```

## 2. Data Minimization Collect and store only necessary data:

- **Event Schema Design:** Include only required fields in event schemas.
- **Field-Level Encryption:** Encrypt only sensitive fields rather than entire events.
- **Retention Policies:** Set appropriate retention periods for different topics.

```

// Example of data minimization in Protobuf schema
message OrderCreated {
    string order_id = 1;
    string user_id = 2; // Store ID, not full user details
    repeated OrderItem items = 3;
    float total_amount = 4;
    string timestamp = 5;

    // Omit unnecessary fields like user's full name, address, etc.
}

```

## 3. Audit Trails Maintain comprehensive audit trails for compliance:

- **Audit Events:** Publish dedicated audit events for sensitive operations.
- **Immutable Audit Logs:** Store audit events in dedicated, immutable topics.

```

// Go example of audit logging
func AuditMiddleware(next message.HandlerFunc) message.HandlerFunc {
    return func(msg *message.Message) ([]*message.Message, error) {
        // Extract context from message
        userID := msg.Metadata.Get("user_id")
        action := msg.Metadata.Get("action")

        // Create audit event
        auditEvent := &AuditEvent{
            UserID:    userID,
            Action:    action,
            Resource:  msg.Metadata.Get("resource"),
            Timestamp: time.Now(),
            Success:   true,
        }
    }
}

```

```

    }

    // Process the message
    result, err := next(msg)

    // Update audit event if there was an error
    if err != nil {
        auditEvent.Success = false
        auditEvent.ErrorMessage = err.Error()
    }

    // Publish audit event
    auditPayload, _ := proto.Marshal(auditEvent)
    auditMsg := message.NewMessage(uuid.New().String(), auditPayload)
    auditPublisher.Publish("audit.events", auditMsg)

    return result, err
}
}

```

## Compliance Frameworks and Controls

Implement controls required by relevant compliance frameworks:

### 1. Access Controls and Least Privilege

- Assign minimal permissions required for each service.
- Regularly review and audit access permissions.
- Implement separation of duties for sensitive operations.

### 2. Data Classification and Handling

- Classify data based on sensitivity (e.g., public, internal, confidential, restricted).
- Apply appropriate controls based on classification.
- Document data flows and storage locations.

### 3. Monitoring and Incident Response

- Monitor for security events and anomalies.
- Establish incident response procedures.
- Conduct regular security testing.

## Secure Development Practices

### Dependency Management

Vulnerabilities in dependencies can compromise your entire system:

- **Regular Updates:** Keep dependencies updated to patch security vulnerabilities.

- **Vulnerability Scanning:** Use tools like OWASP Dependency Check, Snyk, or GitHub's Dependabot.
- **Pinned Versions:** Use specific versions of dependencies to prevent unexpected changes.

```
// Kotlin example with Gradle
dependencies {
    implementation("org.springframework.kafka:spring-kafka:2.8.5") {
        // Exclude vulnerable transitive dependencies if needed
        exclude(group = "org.vulnerable", module = "component")
    }

    // Use Bill of Materials (BOM) for consistent versions
    implementation(platform("org.springframework.boot:spring-boot-dependencies:2.6.3"))
}
```

### Secure Coding Practices

- **Input Validation:** Validate all inputs, including those from internal services.
- **Output Encoding:** Properly encode output to prevent injection attacks.
- **Error Handling:** Avoid leaking sensitive information in error messages.

```
// Go example of input validation
func validateOrderCreatedEvent(event *OrderCreated) error {
    if event.OrderID == "" {
        return errors.New("order ID is required")
    }

    if event.UserID == "" {
        return errors.New("user ID is required")
    }

    if len(event.Items) == 0 {
        return errors.New("order must contain at least one item")
    }

    if event.TotalAmount <= 0 {
        return errors.New("total amount must be positive")
    }

    return nil
}
```

### Secrets Management

Avoid hardcoding secrets in your application code or configuration files:

- **Environment Variables:** Use environment variables for sensitive configuration.
- **Secret Management Tools:** Use tools like HashiCorp Vault, AWS Secrets Manager, or Kubernetes Secrets.

- **Runtime Injection:** Inject secrets at runtime rather than build time.

```
// Kotlin example with Spring Cloud Config and Vault
@Configuration
@EnableConfigurationProperties
class SecurityConfig {

    @Value("${kafka.sasl.username}")
    private lateinit var kafkaUsername: String

    @Value("${kafka.sasl.password}")
    private lateinit var kafkaPassword: String

    @Bean
    fun kafkaProducerConfig(): Map<String, Any> {
        return mapOf(
            CommonClientConfigs.SECURITY_PROTOCOL_CONFIG to "SASL_SSL",
            SaslConfigs.SASL_MECHANISM to "PLAIN",
            SaslConfigs.SASL_JAAS_CONFIG to """
                org.apache.kafka.common.security.plain.PlainLoginModule required
                username="$kafkaUsername"
                password="$kafkaPassword";
            """.trimIndent()
        )
    }
}
```

## Security in a Polyglot Environment

### Cross-Language Security Considerations

Different languages have different security characteristics and vulnerabilities:

#### 1. Go

- **Strengths:** Strong type system, garbage collection, minimal runtime.
- **Considerations:** Manual memory management in some cases, proper error handling.
- **Best Practices:** Use static analysis tools like `gosec`, follow Go security guidelines.

#### 2. Kotlin/JVM

- **Strengths:** JVM security model, strong typing, null safety.
- **Considerations:** JVM vulnerabilities, large attack surface due to extensive libraries.
- **Best Practices:** Keep the JVM updated, use security scanning tools like SpotBugs or SonarQube.

#### 3. Ruby

- **Strengths:** Developer productivity, extensive standard library.

- **Considerations:** Dynamic typing can hide issues, interpreter vulnerabilities.
- **Best Practices:** Use Brakeman for security scanning, follow OWASP Ruby on Rails guidelines.

### Consistent Security Policies

Implement consistent security policies across all services, regardless of language:

- **Centralized Authentication:** Use a single authentication service or identity provider.
- **Standardized Encryption:** Use the same encryption algorithms and key management across services.
- **Unified Logging and Monitoring:** Aggregate logs and metrics from all services for comprehensive security monitoring.

### Conclusion

Security and compliance in Kafka-based, polyglot event-driven systems require a comprehensive approach that addresses authentication, authorization, encryption, and regulatory requirements. By implementing appropriate security controls at each layer—from Kafka configuration to application code—and maintaining consistent security policies across your polyglot environment, you can build systems that protect sensitive data while meeting compliance obligations.

Key takeaways:

1. **Secure Kafka Configuration:** Implement authentication, authorization, and transport encryption for your Kafka cluster.
2. **Protect Sensitive Data:** Use encryption, masking, or tokenization for sensitive data in events.
3. **Comply with Regulations:** Implement mechanisms for data privacy, including the right to erasure and data minimization.
4. **Secure Development:** Follow secure coding practices, manage dependencies, and handle secrets properly.
5. **Address Polyglot Challenges:** Understand the security characteristics of each language and maintain consistent security policies.

By following these principles, you can build secure, compliant event-driven systems that protect your data and maintain the trust of your users and stakeholders.

This concludes our exploration of polyglot event-driven systems with Kafka. Throughout this book, we've covered the principles, implementation details, patterns, and operational aspects of building and running these systems. By combining the strengths of different languages and leveraging Kafka's robust event streaming platform, you can build scalable, resilient, and maintainable systems that meet the demands of modern applications.